

Review Article

Thermal energy storage-centric solar drying with phase change materials: Intelligent optimization via neural and evolutionary regression models

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ABSTRACT

Solar drying offers a sustainable alternative to energy-intensive and environmentally harmful food preservation methods. This review highlights advances in efficiency and reliability achieved through the integration of Phase Change Material (PCM) for thermal energy storage, as well as optimization using neural algorithms optimization and Evolutionary Polynomial Regression (EPR). PCMs absorb surplus heat during peak radiation and release it later, stabilizing drying temperatures, reducing energy demand, shortening drying times, and improving product quality. Studies across different dryer designs confirm these benefits. Machine learning methods, including ANN, SVM, LSTM, and ensemble models provide accurate prediction and optimization of drying performance, surpassing conventional modeling. EPR also demonstrates a strong capability in forecasting outlet temperature and thermal efficiency when PCM parameters are considered. The review further covers solar dryer classifications, PCM selection and placement strategies, and integration with solar collectors under varying conditions. Remaining challenges include limited experimental datasets and the need for advanced computation to model dynamic heat storage. Overall, combining PCMs with intelligent optimization offers a promising pathway to more efficient, resilient, and sustainable solar drying technologies.

1. Introduction

Preserving fruits and vegetables through drying is experiencing a surge in popularity and a growing market demand. This heightened interest stems from the extended shelf life and reduced transportation costs associated with dried produce. Despite the benefits, conventional drying methods often present significant challenges, primarily related to their substantial energy requirements and associated economic drawbacks [1–4]. These traditional approaches can be energy-intensive, leading to high operational costs and raising concerns about their long-term financial viability. Citing these issues, research and development efforts have recently focused on creating more sustainable alternatives. In response to the drawbacks of traditional methods, innovative and environmentally conscious drying technologies have emerged. These newer dryers leverage the readily available and renewable power of solar energy, along with other sustainable resources, to minimize environmental impact and promote energy efficiency. The adoption of

these renewable energy sources significantly reduces reliance on fossil fuels and contributes to a more ecologically sound and economically responsible drying process [5–7].

Solar drying technologies offer a promising avenue for decreasing our reliance on traditional energy sources, thereby contributing significantly to the reduction of both energy consumption and the associated carbon dioxide (CO₂) emissions [8]. These advantages position solar dryers as a valuable tool in furthering sustainable energy initiatives within the framework of future societal development and resource management, as highlighted in existing research [9]. Specifically, the implementation of solar dryers represents a crucial and effective solar energy application, particularly in the context of preserving agricultural produce. A key benefit lies in their ability to mitigate the swelling, spoilage, and subsequent degradation of fruits and vegetables after harvesting [10]. This is especially important considering that such post-harvest losses, largely attributable to inadequate drying and preservation techniques, can account for a substantial proportion, estimated to be between 30 % and 40 %, of the total losses experienced within the

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Nomenclature		PCM	Phase Change Material
<i>Equation symbols</i>		dry	Dry conditions
A	Area (m^2)	wet	Wet conditions
c_p	Specific heat capacity ($\text{J/kg}\cdot\text{K}$)	<i>Abbreviations</i>	
h	Enthalpy, heat transfer coefficient (J/kg , $\text{W/m}^2\text{K}$)	ANN	Artificial Neural Network
k	Thermal conductivity ($\text{W/m}\cdot\text{K}$)	AI	Artificial Intelligence
L	Characteristic length (m)	AT	Ambient Temperature ($^{\circ}\text{C}$)
m	Mass (kg)	CE	Collector Efficiency (%)
$\dot{m}$	Mass flow rate (kg/s)	CFD	Computational Fluid Dynamics
P	Power (W)	CIT	Collector Inlet Temperature ($^{\circ}\text{C}$)
Q	Heat transfer (J)	COT	Collector Outlet Temperature ($^{\circ}\text{C}$)
T	Temperature ($^{\circ}\text{C}$, K)	DCIT	Drying Chamber Inlet Temperature ($^{\circ}\text{C}$)
T_m	Melting temperature of PCM	DCOT	Drying Chamber Outlet Temperature ($^{\circ}\text{C}$)
T_p	Freezing temperature of PCM	DT	Decision Tree
t	Time (s)	EPR	Evolutionary Polynomial Regression
U	Overall heat transfer coefficient ($\text{W/m}^2\text{K}$)	ESE	Energy Storage Element
V	Volume (m^3)	ETSC	Evacuated Tube Solar Collector
x	Moisture content (dry basis) (kg/kg)	FPC/FPSC	Flat Plate Collector/Flat Plate Solar Collector
S	Total solar radiation absorbed by collector (J)	FC	Forced Convection
UL	Collector total heat loss coefficient ($\text{W/m}^2\cdot^{\circ}\text{C}$)	FR	Heat Removal Factor
v	Air velocity (m/s)	IF	Solar Radiation Intensity on Front Side of BIFPVT (W/m^2)
W	Work (J/kg)	IR	Solar Radiation Intensity on Rear Side of BIFPVT (W/m^2)
<i>Greek symbols</i>		kNN	k – Nearest Neighbors
α	Thermal diffusivity (m^2/s)	LCA	Life Cycle Assessment
β	Thermal expansion coefficient ($1/\text{K}$)	LHS	Latent Heat Storage
η	Efficiency (–)	LSTM	Long Short – Term Memory
λ	Latent heat of fusion (J/kg)	MAE	Mean Absolute Error
ρ	Density (kg/m^3)	MC	Moisture Content (%)
ΔT	Temperature difference (K)	MLR	Multiple Linear Regression
Δt	Time step (s)	MSE	Mean Squared Error
μ	Dynamic viscosity ($\text{Pa}\cdot\text{s}$)	NC	Natural Convection
ν	Kinematic viscosity (m^2/s)	PCM	Phase Change Material
ϕ	Volume fraction (–)	PTSC	Parabolic Trough Solar Collector
<i>Subscripts and superscripts</i>		PDSC	Parabolic Dish Solar Collector
amb	Ambient conditions	R^2	Coefficient of Determination
air	Air properties	RF	Random Forest
eff	Effective value	RMSE	Root Mean Squared Error
in	Inlet condition	SAH	Solar Air Heater
out	Outlet condition	SHS	Sensible Heat Storage
melt	Melting point	SVM/SVR	Support Vector Machine/Support Vector Regression
solid	Solid phase	TES	Thermal Energy Storage
liquid	Liquid phase	UTSC	Unglazed Transpired Solar Collector
opt	Optimal value	XGBoost	eXtreme Gradient Boosting
max	Maximum value		
min	Minimum value		

agricultural sector after crops have been gathered [11,12]. Therefore, embracing solar drying offers a tangible strategy for minimizing waste and improving food security.

Solar dryers offer a significant benefit through their ability to be combined with PCM. This integration provides multiple improvements to the drying process and the quality of the final product [7,13]. First, the incorporation of PCM results in a noticeably increased drying efficiency. The presence of these materials allows for a more effective utilization of solar energy, leading to faster and more thorough drying. Secondly, the temperature distribution within the drying chamber becomes uniform when PCM is used. This even temperature prevents localized overheating or under-drying, ensuring consistent results throughout the entire batch [14]. A further positive outcome of using solar dryers with PCM is the extended and enhanced shelf life of the

dried products. The consistent drying and temperature control contribute to reduced spoilage and improved preservation. Finally, and perhaps most importantly, the use of PCM enables the continuous operation of the solar dryer even during periods when direct sunlight is unavailable [15,16]. This is because the PCM stores thermal energy during periods of solar radiation and then releases it gradually as the sunsets or clouds obscure the sky. This stored energy allows the drying process to continue uninterrupted, maintaining a stable drying environment. This is due to the energy depletion by the PCM providing heat when the sun is not shining. PCMs improve thermal energy charging/discharging, boosting dryer performance and efficiency. They enhance solar collectors and PV systems' performance by regulating temperature, improving energy harvesting, and extending component lifespan, leading to sustainable, energy-efficient solutions [17–19]. This technique

has been studied and implemented by numerous researchers; the details are discussed in Table 1.

Accurate analysis and effective optimization of solar drying systems, with a particular focus on solar dryers, necessitate the careful monitoring of a range of environmental parameters throughout the experimental process. These critical parameters include, but are not limited to, the Ambient Temperature (AT) of the surrounding environment, the speed of the ambient airflow, and the intensity of solar radiation received [25,26]. Furthermore, precise measurements of the inlet and outlet temperatures of the heat generation source are essential. This heat generation source often consists of solar air heaters or other types of solar collectors that are directly connected to the dryer unit. Within the dryer chamber itself, it is crucial to obtain detailed data on the temperature, airflow speed, and humidity levels [7,27].

However, the process of acquiring accurate data for these parameters is often complicated by various factors. Data acquisition devices are inherently susceptible to errors, which can compromise the integrity of the measurements. Moreover, systematic characteristics of the experimental setup and instrumentation can also introduce biases and inaccuracies into the collected data. As a result, the parameters required for accurate simulation and reliable analysis of the experimental results cannot always be obtained with sufficient precision [28,29]. To overcome these challenges and ensure the validity of the analysis, the application of neural algorithm techniques becomes highly beneficial. These techniques provide a powerful approach for mitigating the impact of data errors and systematic biases, ultimately leading to a more robust and reliable understanding of the solar drying system's performance. The use of neural algorithms is, therefore, a valuable tool in the analysis and optimization of solar drying processes [30]. Neural algorithms have many subsets, such as Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), Support Vector Regression (SVR), k-Nearest Neighbors (kNN), Artificial Neural Networks (ANN), Feedforward ANN, Backpropagation ANN, Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), eXtreme Gradient Boosting (XGBoost), CatBoost, Multiple Linear Regression (MLR), Evolutionary Polynomial Regression (EPR). In the following sections, we will discuss their performance and application in the solar drying system [31–33].

In recent years, many researchers have used neural algorithms in their research on solar energy. Use in solar ventilation [34–36], solar desalination [37,38], solar collectors [39,40], and solar dryers [41,42]. The recent research on neural algorithms in solar dryers is presented in Table 2.

1.1. Methodology

This review follows a systematic and rigorous approach, drawing on a comprehensive analysis of existing empirical studies and numerical investigations in the field of solar energy systems, with a particular focus on solar dryers. Relevant studies were collected from Scopus, Web of Science, Google Scholar, Science Direct, and Springer, covering the period 2000–2025, resulting in an initial dataset of around 200 articles. A three-level screening strategy (title, abstract, and full-text review) was employed to ensure relevance and quality. Studies focusing on solar drying, PCM integration, and intelligent optimization were included, while unrelated or insufficiently detailed works were excluded, yielding a final set of approximately 175 articles. The selected studies were then classified based on solar dryer types, components, industrial applications, and reported performance outcomes. Research on PCM integration was analyzed for material type, placement, and energy storage strategies, highlighting their impact on thermal regulation and drying efficiency. Additionally, optimization approaches using ANN, SVM, LSTM, EPR, and ensemble models were examined, focusing on the improvement of drying time, temperature control, energy efficiency, and product quality, with comparisons to CFD or conventional modeling approaches where available. An analytical framework was developed by compiling and comparing equations, models, and performance metrics

related to energy, exergy, drying kinetics, PCM behavior, and optimization methods. The key findings, limitations, and research gaps were synthesized to provide insights and recommendations for future studies, particularly in the intelligent optimization of PCM-integrated solar drying systems. The structured framework of this review, including classification criteria, focus areas, observed advantages of neural algorithms, and identified limitations, is illustrated in Fig. 1, providing a clear visual summary of the methodology.

2. Overview of solar dryers

Solar dryers offer a sustainable and effective way to dehydrate various products by carefully harnessing environmental elements such as air movement, humidity levels, and ambient temperature. The controlled environment within a solar dryer provides crucial protection for food items, safeguarding them from potential spoilage, infestation by pests, and the detrimental effects of unforeseen weather events like rain or excessive humidity [50,51]. This drying methodology is specifically designed to preserve the inherent quality of the product being dried, ensuring that the final result meets desired standards for taste, texture, and nutritional value. Furthermore, solar drying is advantageous because it upholds a completely natural drying process, avoiding the need for artificial heat sources or chemical preservatives. This reliance on solar energy and natural air circulation allows for a gentle and gradual reduction of moisture content, preventing damage to delicate compounds and preserving the product's authentic characteristics throughout the dehydration procedure. By utilizing a natural process, the resultant dried product retains its essential properties and flavors [22,52].

2.1. Classification of solar dryers based on airflow mechanism

Solar dryers can be broadly categorized based on their airflow mechanisms into three primary types. These classifications help to understand the different ways solar energy is utilized to remove moisture from agricultural products. The first category is natural convection solar dryers, often abbreviated as NC. These types of dryers rely solely on the natural movement of air, driven by differences in temperature and density, to facilitate the drying process. The buoyancy forces created by the heated air rising through the dryer are what enable the moisture to be carried away from the product being dried. No external mechanical assistance is used to move the air [53,54]. The second major category consists of Forced Convection (FC) solar dryers, which are frequently referred to as FC. In contrast to natural convection dryers, these systems employ forced airflow to accelerate the drying process. Typically, this forced airflow is achieved through the use of a fan, either pushing air into the dryer or using a suction fan to draw air through it. This active movement of air allows for a more controlled and often faster rate of drying compared to relying solely on natural convection [35,36,55–57]. Finally, we have hybrid solar dryers. These dryers represent a combination of the principles used in both natural convection and forced convection systems. A hybrid system leverages both natural airflow, generated by buoyancy, and forced airflow, provided by a fan or similar device, to efficiently dry agricultural products. The integration of both methods allows for a potentially more adaptable and effective drying solution [58–61]. The classification of solar dryers based on the air mechanism is shown in Fig. 2.

Many researchers have used solar dryers to dry agricultural products, including vegetables, fruits, and medicinal plants. A rotary dryer with a thermal storage system was used to dry sumac. The increase in thermal efficiency reached 58.69 % [23]. The use of a PTSC collector as a heat collector connected to a cabinet solar dryer was used to dry apples. Also, a thermal storage system was used in the expansion tank. The maximum drying efficiency reached 20.2 % [62]. A solar dish collector connected to an indirect solar dryer was used to dry mint. Alumina nanofluid was used in this system. The maximum thermal efficiency of the drying

Table 1

Recent research on solar dryers with PCM for the drying process of agricultural products.

References	Dryer type	PCM used	Dried product	Key findings
[2]	Mixed-Mode dryer	Paraffin wax	<i>Baccaurea ramiflora</i>	- The use of PCM reduced drying time by 13 h. - The Moisture Content (MC) of the samples decreased from 84 % to 10 % on a wet basis.
[1]	Indirect solar dryer	Paraffin wax	<i>Melon</i>	- The use of PCM increases the exergy efficiency and sustainability index by 20 % and 47 %, respectively. - Reduces the payback period by 35 %
[20]	Indirect solar dryer	Paraffin wax	<i>Kiwifruit</i>	- The use of PCM reduced the MC from 93 % to 11 %. - Reducing the drying time of kiwifruit to 5.5 h
[21]	Indirect solar dryer	Paraffin wax	<i>Red chilli</i>	- The use of PCM increased the thermal and exergy efficiency to 52.3 % and 2.36 %, respectively. - The drying time of the samples was 29 h.
[13]	Solar cabinet dryer	Paraffin wax	<i>Teucrium polium</i>	- The highest thermal efficiency and overall drying efficiency with PCM reached 62.65 % and 40.86 %, respectively.
[15]	Indirect solar dryer	Paraffin wax	<i>Teucrium polium</i>	- The effect of PCM in reducing the temperature of the drying chamber by 2.5° C. - Reduction of specific energy consumption from 12.25 to 11.04 megajoules per kilogram - Improvement in overall drying efficiency by 3.08 %
[22]	Solar cabinet dryer	Paraffin wax	<i>Red chilli</i>	- PCM improved the system performance, increasing the overall efficiency and drying efficiency by 6.39 % and 13.68 %, respectively.
[6]	Solar tunnel dryer	Lauric acid	<i>Cucumber</i>	- Reduction of drying time by 70.5 % during the drying process using PCM. - Increased drying efficiency by 14.3 % using PCM
[23]	solar rotary dryer	Paraffin wax	<i>Sumac</i>	- The improvement of the dehydration performance in the drying process and the system composition with the integration of PCM was 26.38 %. - The increase in the thermal efficiency of the system with the integration of PCM reached 58.69 %.
[24]	Solar cabinet dryer	Paraffin wax	<i>Ginger</i>	- Reduce drying time by 10 % using PCM - Reduce moisture by 21 % in the drying process

system reached 85 % [63]. An indirect solar drying system with a flat plate solar air heater with PCM and reuse of the hot air return flow was used to dry Oleaster. The overall drying efficiency reached 43.42 % [64]. An indirect solar drying system connected to a flat plate solar air heater was implemented by placing PCM inside the drying chamber on

Table 2

Using neural algorithms in solar dryers to estimate parameters.

References	Dryer type	Type of neural algorithm	Key findings
[43]	Mixed-mode dryer	ANN	- The ANN method achieved optimal drying conditions at a speed of 0.096 m/s, resulting in a thermal efficiency of 33.29 %.
[44]	Concentrated photovoltaic-thermal solar dryer	ANN and SVM	- ANN performs better than SVM in predicting MC with an R ² value of 0.999. ANN was also better in the parameters RMSE and MAE.
[42]	Infrared dryer	ANN	- The ANN model outperformed the Wang and Singh mathematical drying models, achieving an R ² of 99.59 %, x ² of 0.0029, and RMSE of 0.0503.
[45]	Greenhouse dryer	ANN- MLR -DT	- Among the models, ANN had the best prediction with values (R ² > 0.91, RASE <33.5 kgH ₂ O hr ⁻¹) for the greenhouse drying system.
[9]	Solar cabinet dryer	ANN	- The ANN obtained a better R ² value than the Computational Fluid Dynamics (CFD) method, which is 0.971.
[46]	Indirect solar dryer	LSTM	- LSTM estimated the output humidity in the solar dryer chamber with high accuracy, with RMSE = 0.645, MSE = 0.416, MAE = 0.352, and MAPE = 2.555 were calculated.
[47]	Microwave dryer	RF - DT	- The DT method performed better than other methods in estimating parameters, and the highest R ² values (0.96) and the lowest MAPE values (0.03) were obtained.
[10]	Indirect solar dryer	ANN - KNN - SVR - XGBoost	- XGBoost provided the best estimates for dried product MC and collector efficiency in this drying system, achieving the highest R ² (0.999964) and lowest MAE (0.001303), RMSE (0.001882), and MSE (0.000004).
[48]	Solar cabinet dryer	EPR	- EPR accurately estimated collector outlet temperature and thermal efficiency (R ² > 0.98), outperforming CFD (R ² > 0.9432).
[49]	Microwave dryer	ANN - SVR	- ANN performs better in estimating MC than the SVR method and has less error. ANN obtained MSE = 0.000196 and R ² = 0.995.

the inner wall of the chamber to increase the drying efficiency. The maximum drying efficiency reached 38.93 % [16]. Table 3 describes the performance of different types of dryers and provides a description of their components.

2.2. Classification of solar dryers based on the type of solar collectors connected

Solar dryers are usually connected to Solar Air Heater (SAH), Flat Plate Solar Collector (FPSC), Evacuated Tube Solar Collector (ETSC), Parabolic Trough Solar Collector (PTSC), Parabolic Dish Solar Collector

(PDSC), and Unglazed Transpired Solar Collector (UTSC) based on the operating temperature. If PCM material with a high melting temperature is used in the drying system, collectors with high outlet temperature, such as PTSC, PDSC, and ETSC, should be used. Table 4 shows the types of solar collectors connected to the cabinet solar dryer for drying the product, and the drying time and thermal efficiency of drying are examined.

3. Governing equations

3.1. Dryer design equations

Table 5 presents the key equations employed in the design and selection process for all the equipment necessary within a solar cabinet drying system. The design process commences with a crucial initial step: accurately defining both the quantity, or mass, of the fruit intended for drying, and, equally importantly, precisely determining the amount of water content that needs to be evaporated from the fruit to achieve the desired level of dryness. This accurate definition of the initial conditions is fundamental for proper system design.

Following the determination of the fruit quantity and the target water evaporation, another significant factor that must be carefully considered is the set of properties related to the food's physical structure. These properties include the type of tissue composing the food, the

thickness of the individual pieces or slices being dried, and the effective surface area exposed to the drying air. These structural characteristics significantly impact the rate of moisture removal [25].

Having defined these parameters, the next stage involves calculating the total energy required to successfully transform the initial fresh product into the desired dried product. This energy calculation must account for the energy needed to both heat the fruit and vaporize the water contained within it. This value is critical for sizing the various components of the solar drying system. In addition to understanding the food properties and energy requirements, the characteristics of the solar collector itself also need to be carefully determined [23].

Key parameters related to the collector include its surface area, the angle of inclination at which it is positioned, and the heat loss coefficient, which represents the rate at which heat is lost from the collector to the surrounding environment. Accurately determining these collector characteristics is crucial for maximizing the system's efficiency [65,66]. Finally, a comprehensive explanation of the purpose and application of each individual equation listed is clearly illustrated within Table 5. This detailed explanation ensures that the reader can readily understand and apply the equations effectively in the design and selection of solar drying equipment.

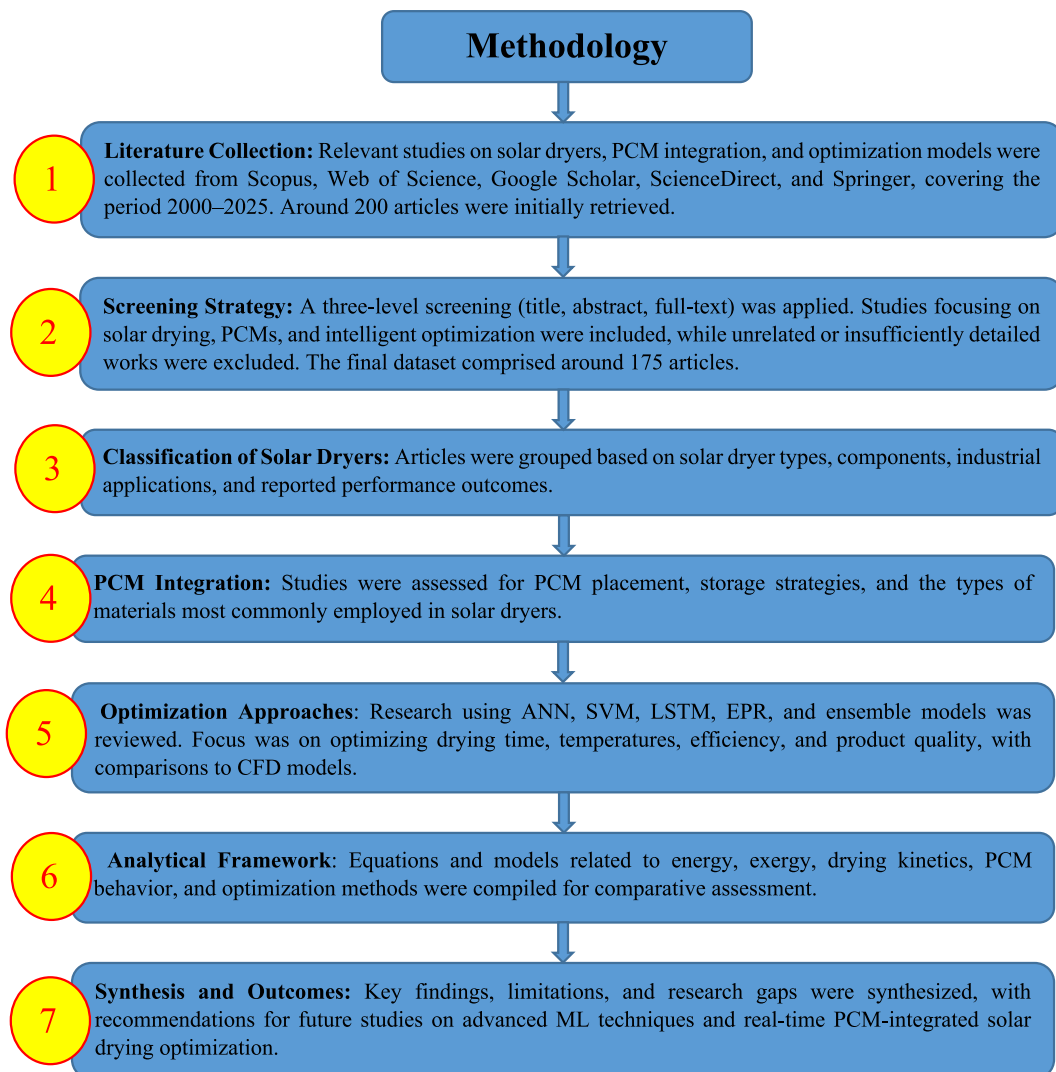


Fig. 1. Methodology in this research.

3.2. Thermal analysis equations

Thermal equations enable the quantification and understanding of heat transfer within systems, allowing for the accurate determination of heat losses from various components [2,7]. Specifically, the Beliss equation [68], is crucial for analyzing solar collector performance, assessing efficiency, and calculating useful energy output. Essentially, a solar collector functions as a heat exchanger, converting solar radiation into thermal energy. The received energy is distributed as useful heat, heat loss, and light loss. Table 6 summarizes key thermal relationships governing solar dryer operation. Exergy equations are also presented for inlet and outlet conditions, enabling a comprehensive exergy transfer analysis. Furthermore, stored energy in PCMs during charging and discharging is considered as a function of fluid temperature, which is crucial for optimizing PCM-based thermal energy storage. Using these equations, it is possible to optimize drying parameters and improve the efficiency of solar drying methods in various applications, facilitating proper storage and value addition of various items. Proper understanding and implementation of these mathematical representations of heat transfer allows for better control and prediction of drying results for various products.

3.3. Mathematical modeling equations

The design and production of solar dryers require equations based on heat and mass balances. These equations are typically expressed as differential equations, integrated with empirical models to characterize the drying kinetics. A prominent example is the Wang and Sing equation, which accurately represents the thin-layer dehydration process. Such equations are essential for enhancing the efficiency of the dryer by simulating the interactions of heat and moisture within the system throughout the drying procedure. Mathematical modeling analyzed bitter melon drying. The drying system's specific energy consumption was 4.44 kWh/kg, with collector and dryer efficiencies of 22 % and 19 %, respectively. The Midili-Kuchuk model best predicted drying parameters [74]. A greenhouse dryer was used to dry the mint plant, and mathematical modeling was used to estimate the parameters. The Wang and Singh model with $R^2 = 0.9944$ performed best [75]. An indirect cabinet solar dryer was used to dry cucumbers. Mathematical modeling was used to analyze the drying parameters. The Midili and Kucuk model

was obtained with very high accuracy, with an R^2 value of >99.5 % [76]. Table 7 presents the mathematical models used to predict drying process parameters.

3.4. Statistical analysis equations

The utilization of Artificial Intelligence (AI) within solar drying systems incorporating PCM necessitates the careful application of statistical analysis equations. These equations are crucial for understanding and optimizing the performance of such hybrid drying systems. In essence, when AI algorithms are employed to control or analyze solar dryers that also integrate PCM for thermal energy storage, a strong foundation in statistical methods becomes indispensable. This is because statistical analysis equations provide the tools to evaluate the effectiveness of the AI algorithms, assess the impact of PCM on the drying process, and ultimately improve the overall efficiency and reliability of the solar dryer. The proper selection and implementation of these statistical techniques allows for rigorous evaluation of experimental data obtained from these advanced solar drying setups, which is critical to validate performance claims and refine system design [10,78]. An ML algorithm was used to predict the MC of red pepper dried using an indirect solar dryer with PCM. Ultimately, the neural algorithm gave a very good estimate, resulting in RMSE = 0.001882, MSE = 0.000004, and $R^2 = 0.999964$ [10]. An indirect solar drying system was used to dry bananas. The XGBoost neural algorithm was used to predict the specific energy consumption parameter, which yielded $R^2 = 0.9957$ and MSE = 0.0747 [79]. Table 8 presents equations for statistically predicting solar dryer performance.

4. An overview of phase change material

Thermal Energy Storage (TES) systems can be defined as mechanisms that serve as temporary repositories or preservers of thermal energy. Essentially, they act as holders or savers of heat or cold, enabling its utilization at a later time. These systems are implemented across a wide range of environmental conditions, making them highly versatile and adaptable to diverse operational needs [7,81]. The concept of TES is far from recent; it boasts a long history of practical application. For many centuries, TES principles have been employed in a variety of agricultural and industrial endeavors. Historical examples include its use in drying

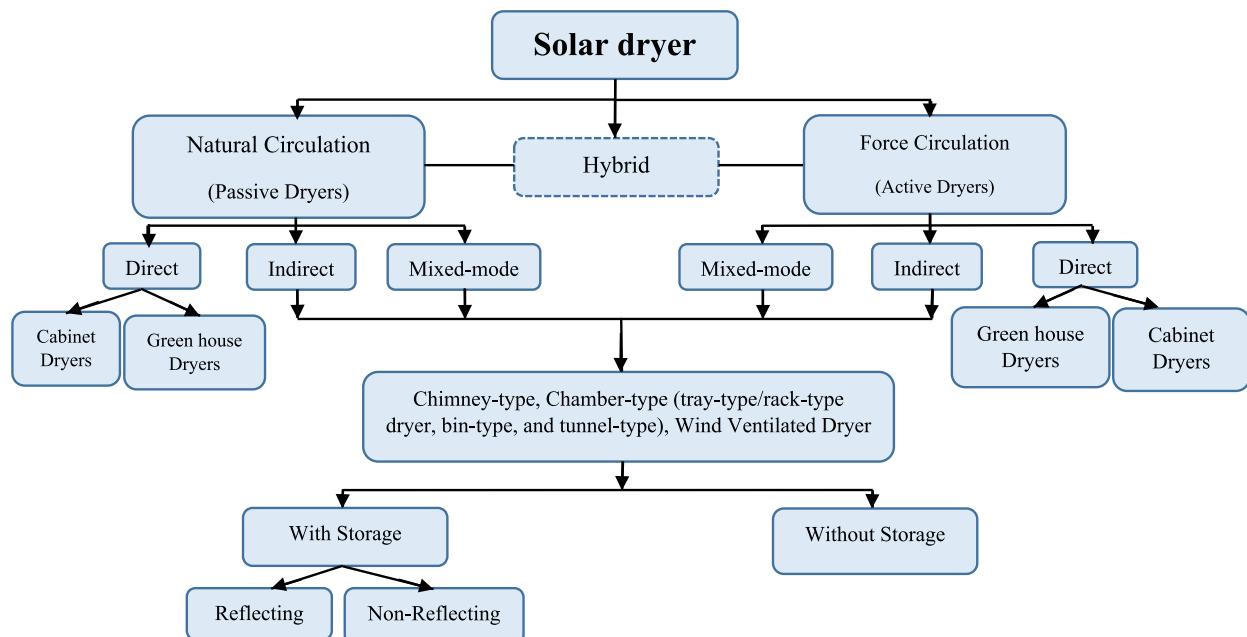
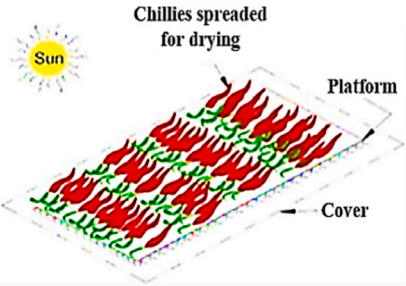
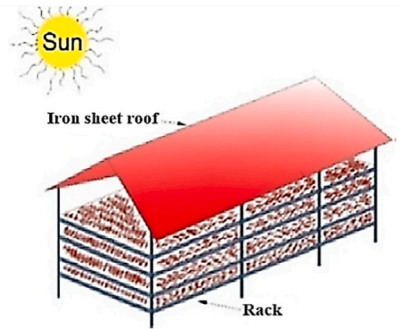
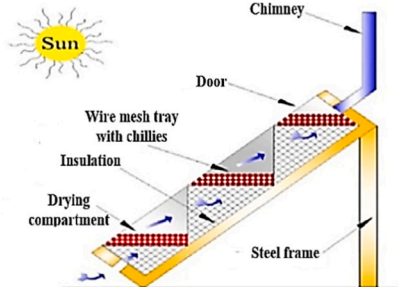
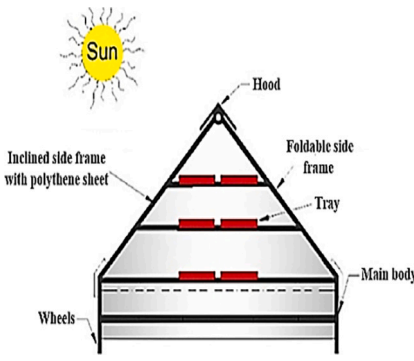
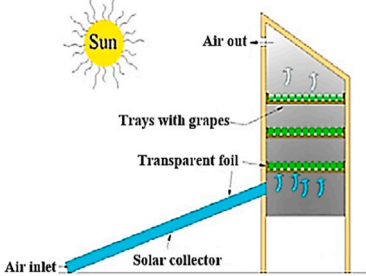
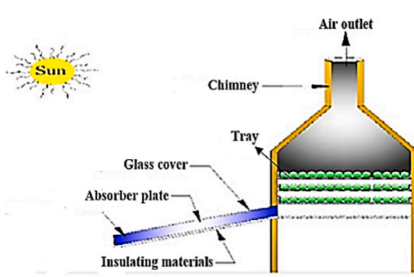


Fig. 2. Classification of solar dryers based on airflow mechanism.

Table 3
The performance of different types of dryers and descriptions of their components [7].

Dryer type	Function description
Natural sun-drying 	Natural sun-drying uses solar energy to evaporate moisture from materials. This method involves placing items in direct sunlight, where the sun's heat transforms water within the material into vapor, which then dissipates, resulting in drying.
Natural rack or shade drying 	Natural rack or shade drying uses ambient conditions in shaded or outdoor areas to evaporate moisture from items like clothes or produce. Items are placed on racks to maximize airflow for evaporation, relying on temperature, humidity, and wind. Shade drying avoids direct sunlight to prevent damage and uneven drying, offering a simple, energy-efficient alternative.
Staircase solar drying 	Staircase solar drying utilizes tiered surfaces, resembling a staircase, to maximize sunlight exposure and efficiently heat materials for drying. This increased surface area enhances solar energy absorption, accelerating moisture evaporation and reducing the material's water content.
Foldable solar drying	Foldable solar drying represents a method of harnessing solar energy for dehydration, utilizing a system for easy storage and portability. This approach to drying employs the sun's radiant heat within a structure that can be collapsed or folded, enabling

Dryer type	Function description
 Solar dryer with natural ventilation 	convenient deployment and retrieval. The foldable nature of the apparatus allows for space-saving storage when not in use and facilitates transportation to optimal locations for sun exposure, making it a practical solution for various drying needs. A key feature of many such dryers is the incorporation of natural ventilation. This means that the movement of air within the dryer is driven by natural forces, such as temperature differences and wind, rather than relying on fans or other mechanical means. This natural airflow is crucial for removing moisture released from the drying material, ultimately leading to a more efficient and sustainable drying process.
Solar dryer with natural ventilation and chimney 	A solar dryer that utilizes the power of the sun for drying purposes incorporates both natural ventilation and a chimney to enhance its efficiency. This type of dryer leverages solar energy to heat the air inside its structure, which in turn removes moisture from the materials being dried. Crucially, the system also incorporates natural ventilation principles, relying on the natural movement of air, along with a chimney, to facilitate the expulsion of moist air and the intake of fresh, dry air, optimizing the drying process.

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Table 3 (continued)

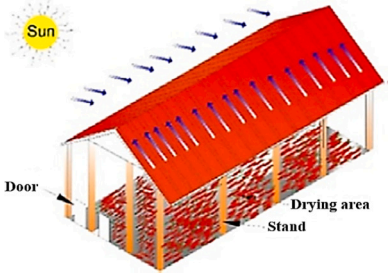
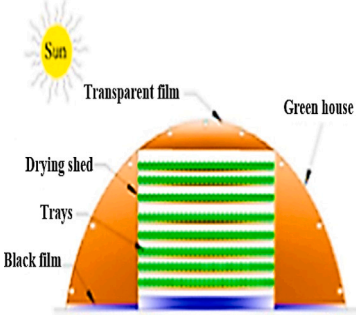
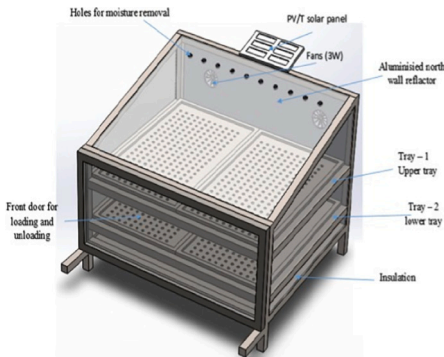
Dryer type	Function description
Roof-integrated solar drying	Roof-integrated solar drying describes a method of drying materials by harnessing solar energy and incorporating the drying system directly into the structure of a building's roof. This approach allows for the utilization of the roof surface as both a solar collector and a drying chamber, maximizing space and energy efficiency. By integrating the drying process with the roof, it avoids the need for separate drying equipment and structures, offering a streamlined and cost-effective solution for certain drying applications. This technique harnesses solar radiation to elevate the temperature inside the greenhouse, thereby reducing the MC of the items being dried. The enclosed space of the greenhouse helps to trap heat and shield the drying materials from external weather conditions such as rain, wind, and dust, which can compromise the drying process and the quality of the final product. Tunnel solar drying is a method of preserving agricultural products or other materials by utilizing solar energy within an enclosed tunnel structure. This drying technique harnesses the power of the sun to evaporate moisture from the substance being dried, leading to a longer shelf life and reduced spoilage. The tunnel design helps to trap solar radiation,
	
Green house solar drying	
	
Solar drying tunnel	
	

Table 3 (continued)

Dryer type	Function description
Solar cabinet dryer	increasing the air temperature inside and accelerating the drying process compared to open-air sun drying. A cabinet solar dryer is a type of drying apparatus that utilizes solar energy as its primary power source. The structure is typically enclosed within a cabinet-like construction, providing a contained environment for the drying process to take place. This design harnesses the sun's heat to effectively remove moisture from various materials placed inside.
	

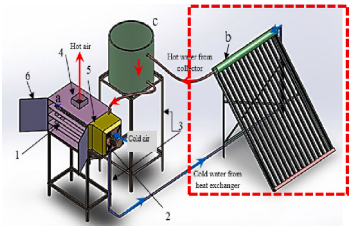
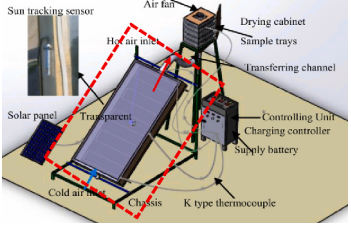
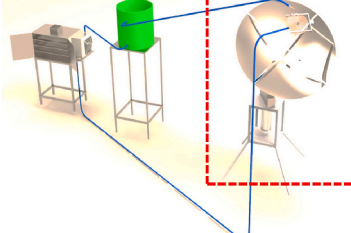
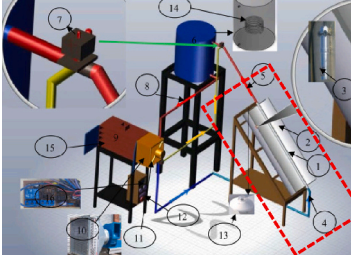
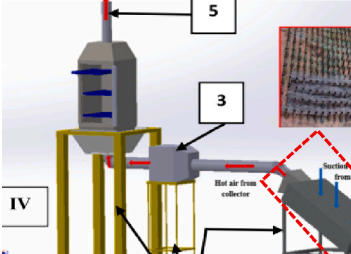
processes, where the stored thermal energy aids in moisture removal, and also in building heating systems, providing a means to regulate indoor temperatures and improve energy efficiency through the storage and release of heat. Therefore, the use of TES is not a novel innovation but rather a long-standing practice with a proven track record in various sectors [7,82]. TES utilizes liquids or solids, insulated to minimize heat loss, by changing the material's internal energy. This change can be sensible heat (temperature increase without phase change), latent heat (phase change at constant temperature), or a combination of both, enabling efficient energy storage [7]. Fig. 3 illustrates the main TES material types, categorized as sensible, latent, or chemical energy storage. Material selection depends heavily on thermophysical properties like specific heat capacity, thermal conductivity, and density, which govern energy absorption, storage, and release [83]. These properties dictate the suitability of each material for specific applications. Thermochemical systems provide TES through reversible chemical reactions, absorbing energy (“charging”) and releasing it (“discharging”) through bond breaking and reforming. The required material quantity is proportional to the endothermic heat of reaction; a higher heat requirement necessitates more material. Reaction kinetics, the rate of energy conversion to stored heat, is also critical for system performance and optimization.

4.1. Sensible heat storage

Sensible Heat Storage (SHS) is a method of storing thermal energy that relies on altering the temperature of a storage medium, which can be either a solid or a liquid. This process leverages the heat capacity of the chosen material to absorb and release energy. The fundamental principle behind SHS involves increasing the material's temperature to store energy and decreasing the temperature to release it. Several factors influence the amount of heat that can be stored using SHS. These key factors include the inherent properties of the storage material itself, the extent of temperature variation experienced by the material during the storage cycle, and the overall quantity or mass of the material being used. As highlighted in the studies by Bal et al. [84] and Srinivasan et al. [85], a comprehensive understanding of these factors is crucial for designing and optimizing SHS systems. Eq. (1) provides a means of

Table 4

Using various types of solar collectors connected to an indirect solar dryer to dry the product.

Type of solar collectors	Image of the solar dryer system	Product	Drying time and drying efficiency
ETSC [3]		Apple	(900–960 min), 39.9 %
FPSC [48]		Tomato	(780–840 min), 25.72 %
PDSC [13]		Teucrium polium	(810–840 min), 40.85 %
PTSC [62]		Apple	(540–630 min), 20.2 %
UTSC [15]		Teucrium polium	(920–960 min), 39.54 %

calculating the thermal energy that is produced by the movement of fluids within the SHS system, allowing for a quantitative analysis of the energy transfer processes involved. This equation typically considers parameters such as the fluid's flow rate, specific heat capacity, and the temperature difference between the inlet and outlet of the storage system.

$$Q = \int_{T_i}^{T_f} mC_p dt = mC_{ap}(T_f - T_i) \quad (1)$$

where C_p is the specific heat capacity (J/kg.K), representing the heat energy required to raise the temperature of 1 kg of a substance by 1 K.

C_{ap} is the average specific heat capacity (J/kg.K) over a temperature range from an initial temperature T_i to a final temperature T_f , both in K. T_i and T_f represent the initial and final temperatures, respectively, of a working fluid undergoing a thermal process.

4.2. Latent heat storage

Latent Heat Storage (LHS) is a technology that stores or releases thermal energy through the phase change of a material at a constant temperature. This process relies on the absorption or desorption of heat during transitions such as melting/solidification, vaporization/conden-

Table 5

Equations used in the initial design of a solar dryer [7,67].

Equations in design	Equation description
$m_w = m_p \left(\frac{m_i - m_f}{100 - m_f} \right)$	Evaporated water content is the water lost from the sample through evaporation. This loss reduces the sample's overall water content.
$E_p = m_w \times L_v$	Evaporation energy is the energy needed to turn liquid water within a product into water vapor. This energy breaks the bonds holding the water molecules together.
$E_a = I_c A_c \eta_c$	Air absorbs radiation, gaining energy. This heats the air.
$E_p = E_a \times t_d$	Evaporation energy = Air's energy gain $\times$ time
$E_a = I_c A_c \eta_c = m_a C_{p_a} \Delta T$	"Heat gain by air" means air's thermal energy increases. This raises the temperature or causes phase changes like evaporation, increasing the air's heat energy.
$m_a = \frac{I_c A_c \eta_c}{C_{p_a} \Delta T}$	Calculation of air mass flow rate in the drying process
$m_a = \rho_a V_a$	Calculating the air speed required in the drying process
$V_a = A \times v_a$	
$V_a = \frac{m_a}{\rho_a}$	
$v_a = \frac{V_a}{A}$	Calculating air velocity based on volumetric flow rate
$A = h \times w$	
$M_{dr} = \frac{m_w}{t_d}$	The average drying speed of the product in the drying process
$I_c A_c \tau \alpha = Q_u + Q_l$	Determining heat losses from a solar air collector requires quantifying the total transmitted and absorbed energy, a key factor in evaluating its thermal performance and efficiency.
$Q_u = m_a C_{p_a} \Delta T$	
$Q_l = U_l A_c \Delta T$	
$F_R m_a C_{p_a} (T_o - T_i) = \frac{K A_c (T_o - T_a)}{t_b}$	Specifying the thickness of the insulation used in the solar collector
$t_b = \frac{K A_c}{F_R m_a C_{p_a}}$	

sation, and sublimation/deposition, allowing for efficient energy storage. The LHS capacity, particularly with PCM [7], depends on the PCM's properties and the specific phase transition.

$$Q = \int_{T_i}^{T_m} m C_p dT + m a_m \Delta h_m + \int_{T_m}^{T_f} m C_p dT \quad (2)$$

$$Q = m [C_{sp} (T_m - T_i) + a_m \Delta h_m + C_{lp} (T_f - T_m)] \quad (3)$$

where a_m is represents the melted fraction of the material. Δh_m is the specific latent heat of fusion (J/kg). C_{sp} is the mean specific heat capacity (KJ/kg K) between the initial temperature T_i and the melting temperature T_m , while C_{lp} is the mean specific heat capacity (KJ/kg K) in the liquid phase between T_m and the final temperature T_f .

4.3. Phase change material

4.3.1. Types of PCMs

A material that releases or absorbs energy during a phase change to provide beneficial heat or cooling is known as a PCM. Materials that undergo phase changes, from liquid to solid or solid to liquid, are frequently employed in a variety of industrial applications. In most cases, the perceptible heat is far less than the energy emitted or absorbed during a phase change from solid to liquid or vice versa. Compared to sensible heat storage, a PCM may store and release a lot of energy by melting and freezing at the phase change temperature [86]. When a material phase changes, heat is either absorbed or released, as seen in Fig. 4. PCMs are therefore referred to as LHS materials. PCMs are a very good way to prevent energy waste and balance some of the energy during peak hours [87].

PCMs are used in various systems such as solar power generation, heat pumps, solar cookers, solar desalination systems, and solar dryers.

Table 6

Equations used for thermal analysis in solar dryer systems.

Reference	Equations in thermal analysis	Equation description
[69]	$S = (\tau \alpha)_{av} I_c$	Average intensity of solar radiation received, considering the absorption and transmission coefficients
[70]	$Q_u = A_c F_R ([I_o(\tau \alpha)] - U_L(T_i - T_a))$	Useful energy absorbed (total) by the solar collector connected to the solar dryer
[70]	$A_c = \frac{W(m_i - m_f) h_{fg}}{F_R(\tau \alpha)_e \int_0^t I_T dt}$	Calculating the area of the collector connected to the dryer
[71]	$F_R = \frac{\dot{m} C_p (T_{fo} - T_{fi})}{A_c [S - U_L(T_{fi} - T)]}$	The heat removal factor (FR) is the ratio of the actual heat transfer to the maximum possible heat transfer through the collector plate.
[71]	$\eta_{collector} = F_R(\tau \alpha) - \frac{F_R U_L (T_i - T_a)}{I_T}$	Calculation of collector thermal efficiency
[3]	$\eta_{ef} = \frac{10^{-3} \times W \times L_v}{\dot{m} C_p (T_{f,o} - T_{f,i}) + E_{elec}}$	Drying efficiency calculation
[13]	$L_v = 2502.54 - 2.309 T_{ca}$	
[13]	$SEC = \frac{Q_s + E_{elec}}{W_o}$	Calculation of specific energy consumption in the drying process
[72]	$Ex_{in} - Ex_u = Ex_{dest}$	Exergy balance in the drying process
[64]	$E_{dest} = Ex_{in} + W - Ex_{u,p}$	Calculation of exergy destruction in the drying process
[64]	$Ex_{in} = \dot{m} c_p \left[(T_o - T_i) - T_a \left(\ln \frac{T_o}{T_i} \right) \right]$	Calculation of the exergy entering the dryer chamber
[64]	$Ex_u = \dot{m} \left[c_p (T_o - T_i) - T_a \left(c_p \ln \left(\frac{T_o}{T_i} \right) - R \ln \left(\frac{\rho_{out}}{\rho_{in}} \right) \right) \right]$	Calculation of the exergy output from the dryer chamber
[64]	$Ex_w = \frac{T_a}{T_i} W_{fan} \cdot W_{fan} = \frac{\dot{m} \times \Delta P}{(\rho \times \eta_{fan})}$	Fan work and exergy created by the fan in the dryer system
[7]	$\eta_f' = \frac{\dot{m} c_p \left[(T_o - T_i) - T_a \left(\ln \frac{T_o}{T_i} \right) \right]}{A_p I_{b,T} \left[1 - \frac{T_a}{T_{sun}} \right]}$	Calculating exergy efficiency in the drying process
[73]	$Q_{pcm} = m_{pcm} c_s (T_{pcm} - T_{melt}) \quad T_{pcm} < T_{melt}$ $Q_{pcm} = 0 \quad T_{pcm} = T_{melt}$ $Q_{pcm} = m_{pcm} c_l (T_{pcm} - T_{melt}) \quad T_{pcm} > T_{melt}$	Calculation of the total energy absorbed by the PCM during the charging or discharging process in the drying process
[3]	$E_{st} = Q_f + Q_{pcm}$	Stored thermal energy (including PCM and fluid used)

Table 7

Mathematical models used in the drying process to predict various parameters [7,77].

Model name	Model description
Newton	$MR = \exp(-kt)$
Page	$MR = \exp(-kt^n)$
Modified page	$MR = \exp(-(kt)^n)$
Henderson and Pabis	$MR = a \exp(-kt)$
Logarithmic	$MR = a \exp(-kt) + b$
Midilli et al	$MR = a \exp(-kt^n) + bt$
Wang and Singh	$MR = 1 + at + bt^2$
Two-term	$MR = a \exp(-k_0 t) + b \exp(-k_1 t)$
Two-term exponential	$MR = a \exp(-kt) + (1-a) \exp(-kat)$
Verma et al.	$MR = a \exp(-kt) + (1-a) \exp(-gt)$

Table 8

Statistical analysis equations for predicting the performance of parameters present in solar dryers.

Reference	Description of evaluation metrics.
[78]	$R^2 = 1 - \frac{\sum_{i=1}^N (o_{exp,i} - o_{pre,i})^2}{\sum_{i=1}^N (o_{exp,i} - o_{pre,i})^2}$
[80]	$MSE = \frac{1}{N} \sum_{i=1}^N (o_{exp,i} - o_{pre,i})^2$
[28]	$MAE = \frac{1}{N} \sum_{i=1}^N o_{exp,i} - o_{pre,i} $
[78]	$RMSE = \left[\frac{1}{N} \sum_{i=1}^N (o_{exp,i} - o_{pre,i})^2 \right]^{0.5}$
[78]	$MAPE = \frac{1}{N} \sum_{i=1}^N \left \frac{o_{exp,i} - o_{pre,i}}{o_{exp,i}} \right \times 100$

The operating temperature for drying agricultural products is typically between 40 and 60 °C [83]. Increasing the temperature beyond the expected range reduces the quality of the dried product [202]. In addition, it has significant environmental benefits [89]. The potential of paraffin wax as a PCM in solar dryers has been evaluated by many researchers. They found that PCMs can store excess energy from the sun and discharge this energy when needed [90]. Various parameters, such as heat transfer characteristics, inlet and outlet temperatures, and the effect of air velocity during energy storage and discharge, as well as the quality of the dried product with and without PCM, were studied. Some

of them are mentioned below.

Solid-liquid PCMs, which include organic, inorganic, and eutectic variations, are regarded as one of the most prevalent and effective types of PCMs for TES systems [91]. The most popular and widely used PCMs over the past three decades have been paraffin waxes, salt hydrates, fatty acids, and different organic and inorganic eutectic compounds, according to research and performance analysis of this type of phase change material [92–94]. The majority of thermal storage methods use paraffins, which are high molecular weight hydrocarbons that are waxy at room temperature. Straight-chain hydrocarbons make up paraffins. The quantity of carbon atoms in the material structure has a direct impact on the melting point of paraffin, which range from 6 to 80 °C. Compared to pure paraffins, paraffin waxes have a combination of hydrocarbon molecules with varying carbon numbers, lower melting points, and weaker latent heats. Paraffin waxes are frequently regarded as inferior PCMs. Because of its qualities, which include stability at room temperature and the ability to be employed in any material without corroding, paraffin waxes have been especially appealing as PCM. Paraffin waxes are non-corrosive, safe, dependable, and have known thermophysical characteristics. They are also reasonably priced. They have a larger heat storage capacity, are chemically inert, exhibit minimal volume change during melting, and have little vapor in the melting state [95]. The high heat of fusion, compatibility with the majority of materials, non-reactivity with the majority of encapsulating materials, and chemical stability after numerous heating and freezing cycles are further benefits of paraffin.

4.3.2. PCM in thermal systems

The incorporation of a TES system within a solar dryer represents an enhancement to its operational capabilities. Essentially, this means equipping the solar drying apparatus with a mechanism for accumulating and retaining heat. The term TES system, as it pertains to a solar dryer, refers to a technology used to store thermal energy harvested from solar radiation and then release this stored energy for use in the drying process when direct sunlight is insufficient or unavailable. One significant benefit offered by solar dryers is the reduction in the amount of time required for drying, representing a considerable time saving. Furthermore, solar drying systems offer improved efficiency and higher levels of productivity in comparison to conventional, more traditional

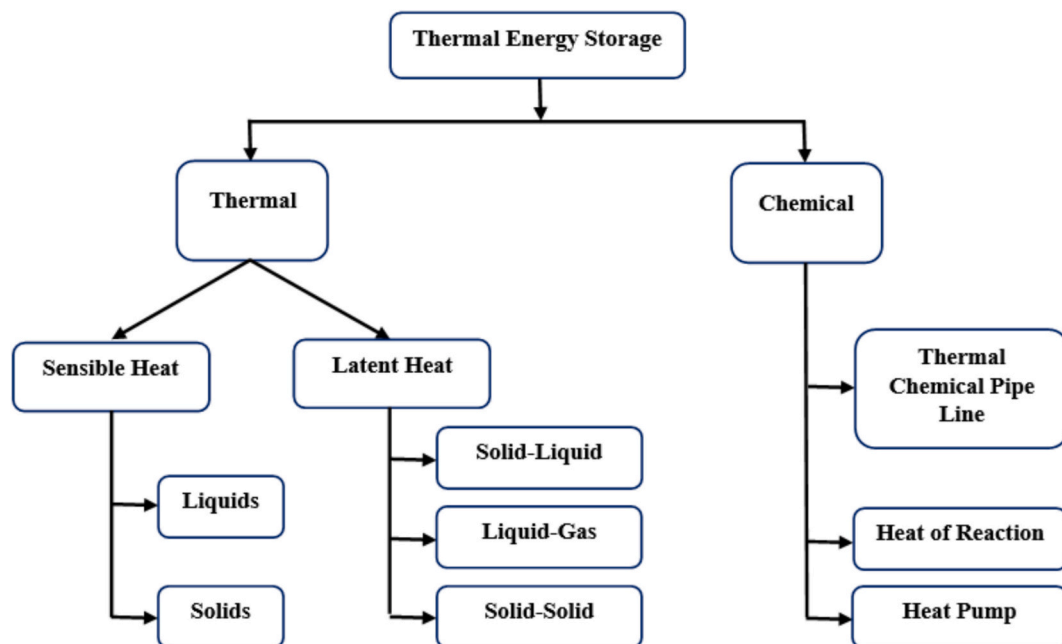


Fig. 3. Review of different types of thermal storage methods with consideration for solar systems [7].

drying techniques. This makes solar dryers a more effective solution for preserving materials. An additional enhancement to solar drying technology involves incorporating thermal accumulators, with PCM being a prominent example, within the drying system. These accumulators serve to extend the duration of the drying process, enabling a more consistent and thorough removal of moisture over a longer period. The inclusion of PCMs allows for more continuous drying, even when solar irradiance is fluctuating or limited. Based on the good performance of using PCM in solar dryers, various studies have been conducted.

Using CFD Fluent, Sandali et al. [96] numerically simulated a solar collector for air heating requirements using phase-change materials. In this study, thermal energy from latent heat was stored in a layer of phase-changing material. According to the study's findings, placing a 1 cm thick PCM beneath the absorber plate caused the output temperature to rise by at least 10°C after sunset and extended the daily energy extraction operation from the collector by up to 2 h. The Taguchi method was used in a study to optimize thermal energy based on phase transition materials in a solar collector. Time, ambient temperature, sun radiation, and phase-changing materials (coconut oil, wax, and grease) were the optimization's input parameters. According to the study's findings, performance was enhanced when PCM was present as opposed to when it wasn't [97]. In tropical residential and commercial settings, phase-change materials were employed in a flat plate solar collector. The purpose of this study was to see whether PCM could be used in both the home and commercial sectors. Using Lauric acid that was 32 mm thick, the maximum temperature ever recorded was 95°C . The internal rate of return served as a sustainability indicator for investments in the economic study. According to reports, the maximum internal rate of return for the industrial and residential sectors was 17.2 % and 22 %, respectively. These materials' integration enhances the system's performance and enables their use in flat plate solar collectors [98]. Badiei et al. [99] used a numerical model to analyze the thermal and fluid behavior of flat plate solar collectors and examined how incorporating fins and PCM could increase performance. The findings show that the PCM system can supply hot water for a longer duration during discharge, despite having a lower morning outlet temperature. Additionally, on a summer day, the average collector efficiency (CE) for PCM with a minimum melting temperature rose from 33 % to 46 %. Additionally, the storage capacity was enhanced by the use and integration of fins, particularly in PCM with a higher melting temperature. The results of the experimental and numerical study on the solar collector using PCM as heat storage showed that the maximum outlet water temperature reached by using PCMs in the receiver tube was roughly 16.80 % and 14.86 % higher than that of the simple tube for July 21 and October 21, respectively. When the receiver with a phase change concentric tube was used on both days, the parabolic trough solar collector's daily thermal efficiency rose by 6.56 and 8.32 %, respectively. For water inlet temperatures of 30°C and 40°C , respectively, PCM demonstrated good thermal stability for two and one hour and forty minutes [100]. PCM was employed in the receiver tube of a study to improve the thermal

efficiency of a parabolic trough solar collector. Inside the focal tube, the PCM was arranged as a 6 mm spiral tube with a 7 cm pitch. The thermal efficiency increased by 3.05 % with the introduction of phase transition materials [101].

4.3.3. PCM in the solar dryers

The other way to improve the performance of the solar dryers is to use PCM inside or outside of the drying system. In an indirect solar dryer to decrease the thermal shocks created by the solar collector during drying times, PCM was used inside the drying cabinet (Fig. 5). PCM was poured into the copper tubes and located below the drying trays under various conditions. The best drying performance of 38.95 % was obtained at the air velocity of 1 m/s when the PCM tubes were located below the first tray [16].

In another solar cabinet drying system, PCM tubes was located vertically through the drying cabinet (Fig. 6). The results showed that using PCM enhanced the drying performance of the system during banana drying process. Thus, the exergy efficiency improved up to 28.1 % [102].

In a solar dryer PCM was considered as TES, to enhance the drying efficiency of a rotary solar dryer. The spiral copper tube with paraffin wax was used inside of the drum (Fig. 7). The result shows an improvement of 37.26 % of drying efficiency due to use TES and recycling system. The best condition for PCM was obtained using computational fluid dynamic method [23].

In a solar dryer, the overall drying efficiency improved at least 2.32 % due to the use of PCM, recycling mechanism, and nanofluid (Al_2O_3). PCM was located around the section of the drying cabinet in a thick layer of paraffin, and Rhubarb slices were placed in the trays (Fig. 8a). Moreover, the findings showed an improvement of exergy efficiency (35.4 % - 61.3 %) [8]. In a cabinet solar dryer, PCM was melted and poured into a grooved aluminum container, and the lid was closed (Fig. 8b). The average collector efficiency using the heat accumulator was 26.11 % and the drying time of the product was 7 h earlier. The cabinet enclosure provided uniform heat distribution for drying the product.

Additionally, PCMs are increasingly adopted in solar drying systems to buffer thermal fluctuations, store excess heat during irradiation peaks, and release it when solar input wanes. These functions contribute to a more stable drying environment, improved energy efficiency, shortened drying time, and enhanced product quality. Table 9 presents recent studies, detailing PCM types, dryer configurations, integration strategies, and observed performance outcomes. Emphasis is placed on energy-storage effectiveness and dryer resilience under real-world conditions.

The charging and discharging behavior of the PCM in a mixed-mode solar dryer running at various air recycling ratios is shown in Fig. 9(a). For recycling ratios of 100 %, 50 %, and 0 %, the PCM charging periods lasted from 09:00 a.m. until approximately 2:30 p.m., 2:00 p.m., and 1:30 p.m., respectively, storing thermal energy in the ranges of 0.033–0.161 kWh, 0.030–0.145 kWh, and 0.001–0.131 kWh. Discharging occurred thereafter until 06:00 p.m., releasing 0.051–0.237 kWh, 0.057–0.182 kWh, and 0.019–0.123 kWh, respectively, to compensate for the energy deficit. Notably, a reduction in the air recycling ratio enhanced solar collector efficiency, achieving values of 55.27 %, 60.59 %, and 66.52 % for 100 %, 50 %, and 0 % recycling, respectively, in both PCM-integrated and non-PCM configurations [104]. A study on solar air dryers equipped with flat plate collectors explored the combined effects of a graphene–alumina hybrid nanofluid and palmitic acid PCM. The hybrid nanofluid, prepared in a 1:3 ratio and circulated at a constant flow rate of 7 L/min, was tested at concentrations of 0.5, 1.0, and 1.5 vol%. The integration of PCM with the nanofluid significantly enhanced thermal storage and release capabilities, ensuring a more stable drying temperature and prolonged heat availability after sunset. At the optimal 1.5 vol% concentration, the system demonstrated superior thermal conductivity ($0.75 \text{ W/m}\cdot\text{K}$), improved heat transfer rate

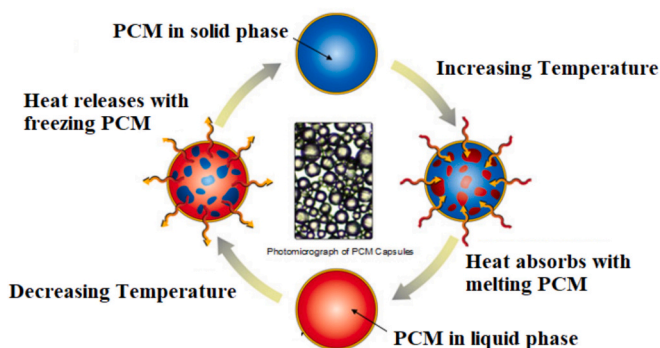


Fig. 4. Phase change process in PCMs [88].

(345.5 W) as shown in Fig. 9(b), reduced heat loss (58.7 W), higher thermal and exergy efficiencies (62.5 % and 20.7 %, respectively), and a dryer efficiency of 46.8 %. These results markedly outperformed a conventional solar dryer using ethylene glycol and palmitic acid PCM, highlighting the synergistic effect of hybrid nanofluids and PCMs in improving overall dryer performance [105].

Fig. 9(c) illustrates the influence of PCM layer thickness on the thermal, electrical, and exergy outputs of a PVT air collector system. The analysis reveals that thinner PCM layers enable superior thermal performance, as they facilitate faster heat transfer and more responsive thermal behavior. A PCM thickness of 0.005 m yields the highest thermal energy output (151 Wh), outperforming thicker layers due to reduced thermal resistance. Conversely, increasing the PCM thickness results in a progressive decline in thermal energy output, with layers of 0.02 m and 0.03 m producing only 75 Wh and 55 Wh, respectively [59]. This trend highlights the importance of optimizing PCM geometry to balance storage capacity with efficient energy extraction. For the configuration with PCM tubes in the plenum chamber, the daytime air temperature remains lower than in the other cases, with maximum values of 62 °C for the system without storage, 61.5 °C with PCM under the absorber, and 53.7 °C with PCM in the plenum chamber. The lower peak temperature in the plenum chamber case is due to reduced direct solar exposure on the storage medium. However, after 8 p.m., the plenum chamber PCM continues releasing heat for a longer duration, extending the useful thermal period. Fig. 9(d) further reveals that PCM under the absorber completes melting and solidification more rapidly, whereas the plenum-mounted PCM undergoes slower phase change, with incomplete melting during daytime operation due to lower thermal exposure [106].

During mango pulp drying, temperature retention varied significantly with PCM type and quantity. Without PCM, the dryer maintained temperatures above 30 °C for 6 h and above 50 °C for 7 h, whereas with 200 g micro-PCM, these durations extended to 6 h above 40 °C and over 5 h above 60 °C. Fig. 9(e) shows the temperature variation for mango pulp drying in different PCMs of the solar dryer. Increasing PCM mass improved high-temperature retention and accelerated drying, with reductions in drying time reaching up to 63 % compared to the PCM-free case [109]. Experimental drying of banana slices was conducted using paraffin wax-filled copper tubes (176 g in PCM1 and 494 g in PCM2) positioned at the solar dryer's inner corners. Fig. 9(f) presents the charging and discharging of PCM in the solar dryer with two different masses. The PCM charged steadily during peak solar hours, storing up to 3.52 kJ, and discharged in the late evening, releasing up to 4.73 kJ, thus extending heat availability beyond sunlight hours. Incorporating PCM improved average exergetic efficiency from 21 % (without PCM) to 28 %

(PCM2), alongside enhanced sustainability and reduced environmental impact [102].

5. Overview of neural algorithms and evolutionary polynomial regression

The integration of scientific computation with machine learning has catalyzed the creation of algorithms aimed at data analysis and outcome forecasting, resulting in models that are both more sophisticated and precise. As a result, data-driven approaches are gaining traction in various scientific and engineering fields. In the context of drying applications, several key strategies are especially pertinent for effective modeling and optimization. This document outlines a range of these fundamental methodologies to offer a thorough overview of the techniques essential for addressing challenges in the drying process. Although this selection is not comprehensive, it provides a robust basis for predictive modeling in drying [10,31,110]. There are a few basic methods that we consider in drying applications, presented in Fig. 10.

5.1. Decision tree

AI has various branches, and one of them is machine learning. A DT is one of the most popular methods in supervised machine learning. Over the years, the algorithm has been enhanced significantly and is now optimized even further for data analysis [111]. A DT is an artificial learning algorithm that splits data into small groups for the purpose of classifying or making numeric predictions (Regression). The model divides data in step-by-step order based on their attributes to generate groups of smaller sizes to make it easy for classification [31]. What a decision tree does is that it first examines the input data and divides it into subsets based on their characteristics. This is carried out recursively until the final subsets are sufficiently pure to allow accurate predictions. Fig. 11 shows the operation of a decision tree and its stages of completion.

The DT algorithm has been used by researchers. For example, Hürdoğan et al. [112], conducted an experimental study with a new chamber solar dryer for drying peanuts. They compared the experimental results with MLR, DT, and the SVM algorithms, and the best algorithm for estimating the experimental data was the DT method with an R2 value of 0.9972. Demirpolat [113], investigated drying process modeling and product MC calculation, comparing a decision tree algorithm with the Midilli numerical model. The decision tree outperformed the Midilli model, predicting the moisture rate parameter with an RMSE error 0.03 units lower. [114], experimentally investigated a greenhouse solar dryer for grapes, using MLP and DT neural algorithms to model the

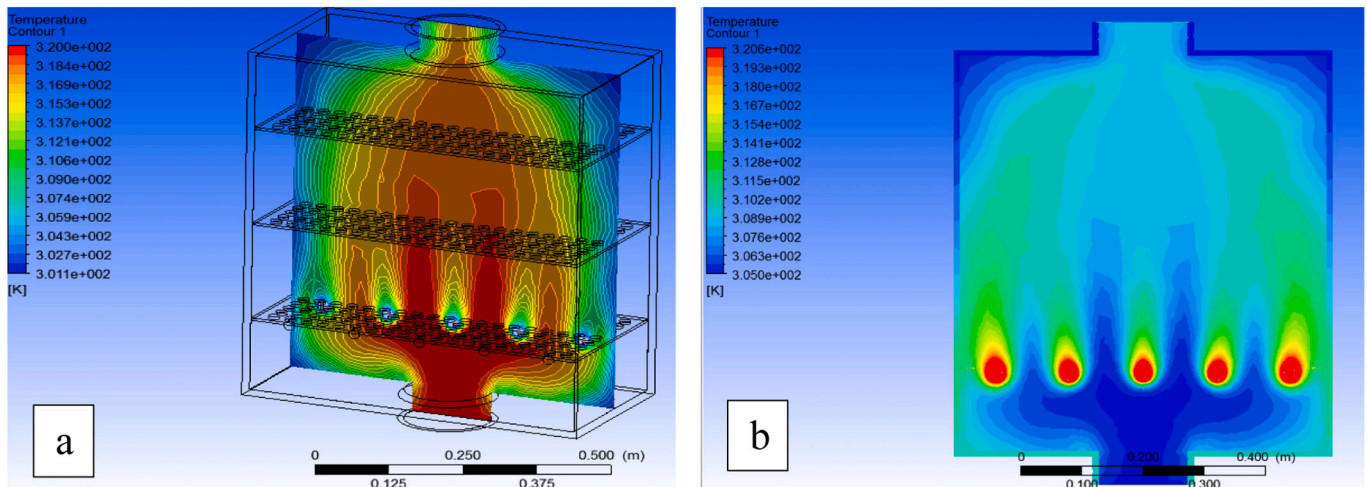


Fig. 5. The thermal contours for the drying cabinet with PCM for a) charging and b) discharging processes [16].

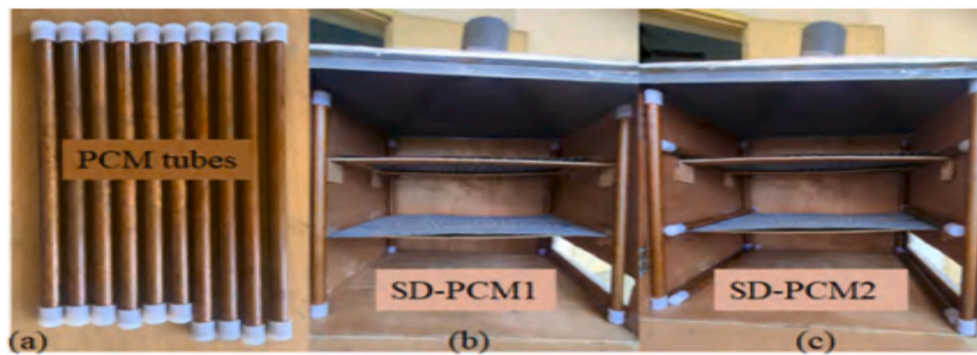


Fig. 6. (a) Copper pipes filled with PCM, (b) placing 4 vertical copper pipes in the cabinet dryer compartment and (c) placing 10 copper pipes in the cabinet compartment [102].

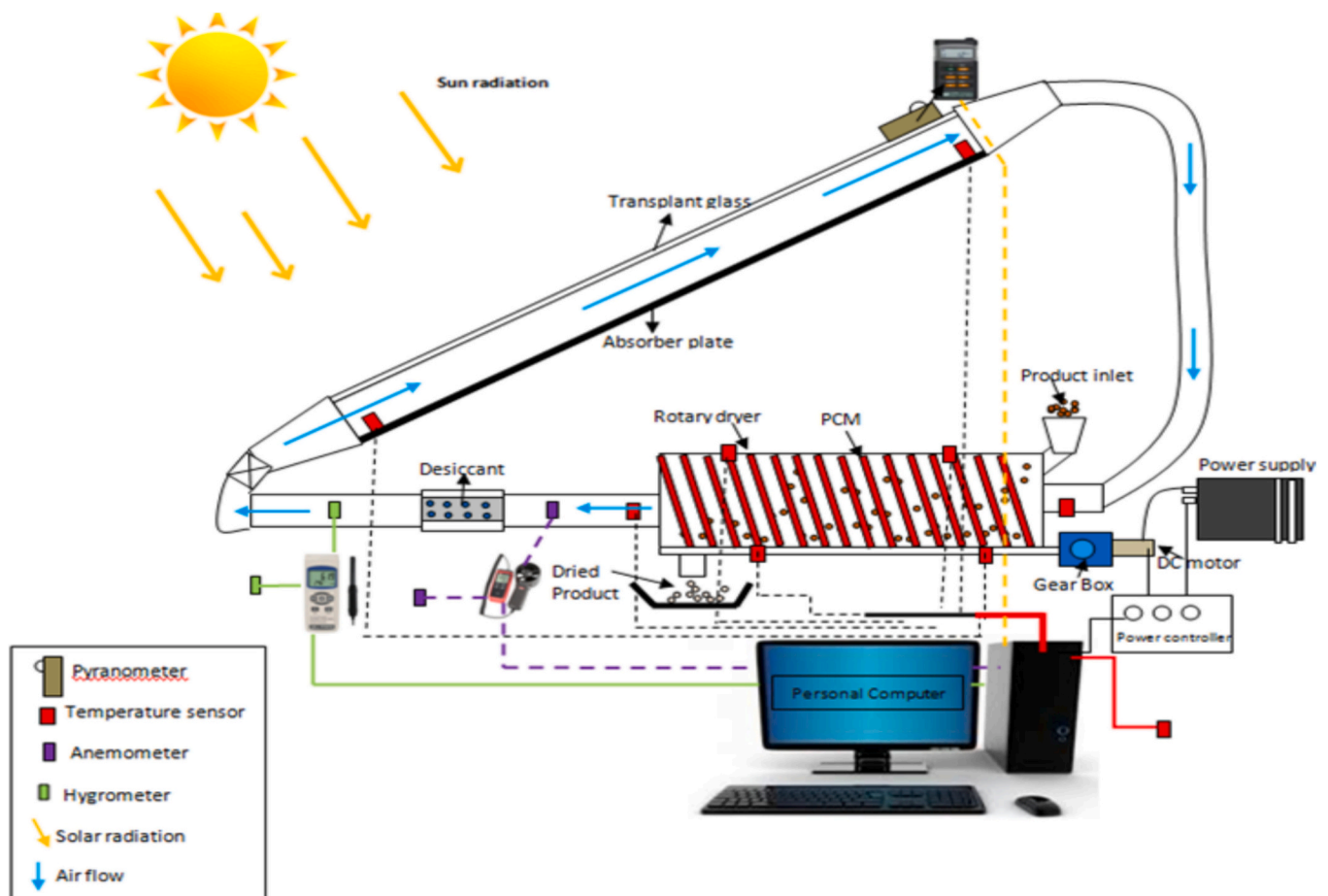


Fig. 7. Rotary solar dryer system with return flow mechanism, with heat storage inside a copper tube inside the drum chamber [23].

heat transfer coefficient. Results showed the decision tree performed significantly well (RMSE = 0.0815).

Other researchers have used decision trees to predict energy consumption. [115], A DT, along with meteorological variables for solar power forecasting and was implemented in MATLAB software Simulink. [116], Compared the performance of the decision tree with three other algorithms (ANNs, SVR, and RF) for predicting energy consumption in open pit mines. The results showed that the RF performed better than the decision tree. While the ANN method is discussed in a subsequent section, it's important to highlight a key advantage of decision trees right now. One significant strength of the decision tree approach lies in its capacity to enhance its performance through the integration of physics-

based machine learning techniques. Specifically, leveraging insights from physics can lead to improvements across several key areas. These include boosting the efficiency of the DT algorithm, increasing the accuracy of its predictions, and making the model's decision-making process more transparent and easily interpretable. This ability to leverage physics-based approaches to optimize the decision tree is supported by evidence in the research literature, as indicated in reference [117].

5.2. Random Forest

RF is a highly favored and widely used ML algorithm, renowned for its power and adaptability within supervised learning paradigms. It is a

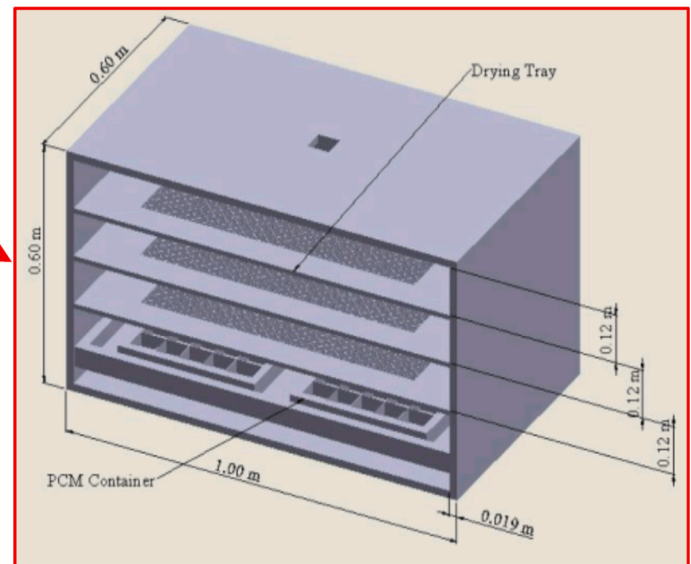
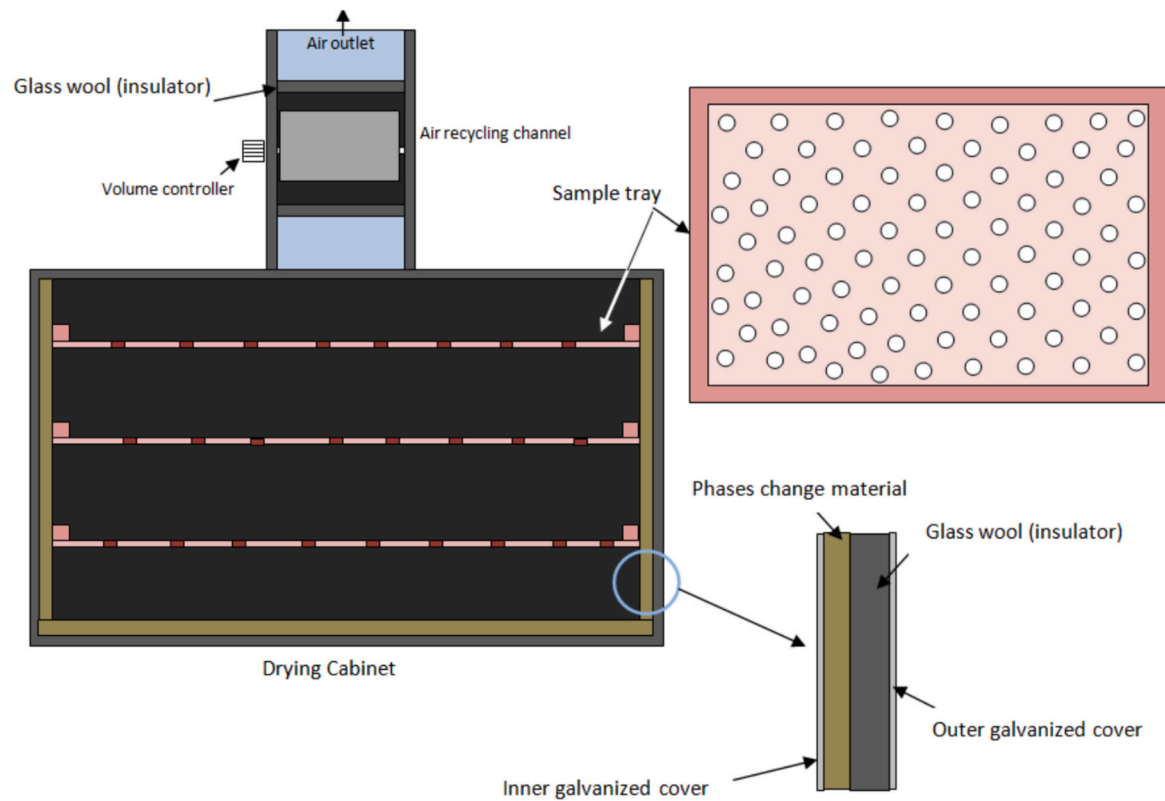


Fig. 8. (a) PCM was located around the section of the drying cabinet to dry Rhubarb [8], (b) solar drying system with PCM placement in a grooved aluminum container [103].

method employed to construct a resilient and consistent model specifically designed to enhance the accuracy of our predictions, striving for optimal efficiency in the process. Consequently, RF is classified and understood as a member of the supervised learning branch within the broader field of machine learning [118,119]. The strength of the RF algorithm lies in its ability to handle diverse datasets and prediction tasks effectively. Its versatile nature allows it to be applied across various domains, making it a valuable tool for data scientists and ML practitioners. The core objective of RF is to develop a model that provides accurate predictions and remains stable and robust in the face of variations in the input data. This focus on both performance and

reliability contributes to its popularity and widespread adoption. By operating under supervision, the algorithm learns from labeled data, refining its predictive capabilities and ensuring its classification as a supervised learning technique [28,31,120]. RF, an ensemble learning algorithm, combines multiple DTs to improve accuracy and robustness. By training each tree on different data subsets and features, it creates a “forest” that outperforms single DTs. Versatile and applicable to both classification (assigning data to categories) and regression (predicting numerical values), Random Forest's functionality is illustrated in Fig. 12.

The performance of random war, according to the diagram: RF uses this technique (Bagging - Bootstrap Aggregating) to create multiple DTs

Table 9

Recent studies on solar dryers with PCM effect and their outcomes.

Reference	PCM	Solar dryer	Quantitative effect of PCM	Key findings
[104]	Paraffin wax 52 °C	Mixed-mode solar dryer with various air recycling ratios	Incorporating PCM in the mixed-mode solar dryer effectively stores excess midday thermal energy and releases it during later hours, sustaining drying performance and reducing energy shortfalls after sunset (Fig. 10a). The dryer efficiency was 18.51 %.	Reducing the air recycling ratio shortens PCM charging time, increases solar collector efficiency by up to 11.25 %, and enhances the PCM's ability to offset evening energy deficits. The dryer completes drying of coffee beans within 9 h and has a faster payback period of 1.56 years.
[105]	Palmitic acid	Solar air dryer with different vol % of nanofluid with PCM	Inclusion of palmitic acid PCM with 1.5 vol% graphene-alumina hybrid nanofluid increased thermal conductivity to 0.75 W/m·K, while improving thermal and exergy efficiencies by up to 62.5 % and 20.7 %, respectively.	Hybrid nanofluids combined with PCM provide higher heat retention, reduce losses by nearly 59 W, and significantly enhance post-sunset drying capability compared to conventional PCM-based systems.
[59]	RT44HC	Solar air collector with different thickness	A 0.005 m PCM layer yields the highest thermal output of 151 Wh, outperforming thicker layers (0.03 m produces only 55 Wh). The system has a dryer efficiency of 15.23 % and reduced the MC by 26.6 %.	Thinner PCM layers (<0.01 m) significantly improve solar-to-thermal energy conversion in PVT air collectors by enabling faster heat transfer and higher thermal efficiency.
[106]	Paraffin wax 55.68 °C	ISD with PCM installed in different locations	Maximum air temperature drops from 62 °C (no storage) and 61.5 °C (PCM under absorber) to 53.7 °C when PCM is placed in the plenum chamber. The PCM incorporation increases the cost by 38.87 % when compared to without PCM.	Locating PCM in the plenum chamber prolongs post-sunset heat release but reduces peak daytime temperature due to slower and incomplete melting.
[107]	C ₂₃ H ₄₈	Solar air dryer with 50 % and 100 % filled PCM	Reducing PCM volume from 100 % to 50 % increased daily heat collection efficiency from 56.44 % to 63.09 %, representing a 6.65 % improvement. The	Optimizing PCM volume to 50 % can improve efficiency, reduce costs, and enhance commercial viability without compromising

Table 9 (continued)

Reference	PCM	Solar dryer	Quantitative effect of PCM	Key findings
[108]	OM 49	ISD with and without PCM	100 % filled PCM has a shorter cost recovery period. Incorporating PCM reduced final MC from 7.83 % to 6.67 %, increased drying rate from 12.16 to 13.37 kg/h, and improved drying efficiency (51.84 %) by 1.51 % than without PCM.	thermal storage performance. PCM integration enabled steadier moisture loss, reduced shrinkage (1.94 % vs. 3.35 %), and extended drying time for enhanced thermal performance and nutrient retention.
[109]	Paraffin wax	Solar dryer with macro (56 °C) and micro (56.8 °C) PCM	Use of 200 g micro-PCM in the solar dryer maintained temperatures above 40 °C for 360 min and above 60 °C for 310 min, resulting in a 63 % reduction in drying time compared to the dryer without PCM.	Higher PCM mass and optimized placement significantly extend high-temperature duration in solar drying, directly reducing overall drying time and improving product quality.
[102]	Paraffin wax	Solar dryer with copper tubes filled 176 g (PCM 1) and 494 g (PCM 2)	PCM 2 achieved an exergetic efficiency of 28 %, compared to 21 % without PCM, with a maximum cumulative heat release of 4.73 kJ. The drying time was 9 h in the PCM 1 system, whereas in PCM 2 it took 15 h.	Integrating PCM significantly improved energy retention and sustainability, enabling effective drying during low-radiation periods.

that are more effective and robust. The training data is divided into several random subsets (some samples may be selected multiple times) and then a separate DT will be trained for each subset. In addition, when the model wants to make a prediction, each of the trees presents the results recorded in the method it used and finally, the results of all of them are combined, and the model of their combination will depend on the type of problem.

In the construction of each DT within the algorithm, a specific procedure is implemented at every node. Rather than assessing all available features to determine a split, the algorithm employs a method of random feature selection. This approach entails that for each node being evaluated, a subset of features is intentionally chosen based on a random number from the complete set of possible features. Once this random subset is established, the algorithm focuses solely on these selected features. From this restricted set, it identifies and determines the optimal split point, disregarding all other potentially useful features that are not part of the random selection. The pertinent steps are illustrated in Fig. 13.

The random forest algorithm has found application across numerous domains of research, and the study of solar systems is no exception. Within this broader field, researchers have specifically utilized the random forest algorithm in a diverse array of approaches focused on solar energy systems. One particular area where this machine learning technique has proven valuable is in the context of solar dryers. In these applications, the algorithm is employed to analyze and model various

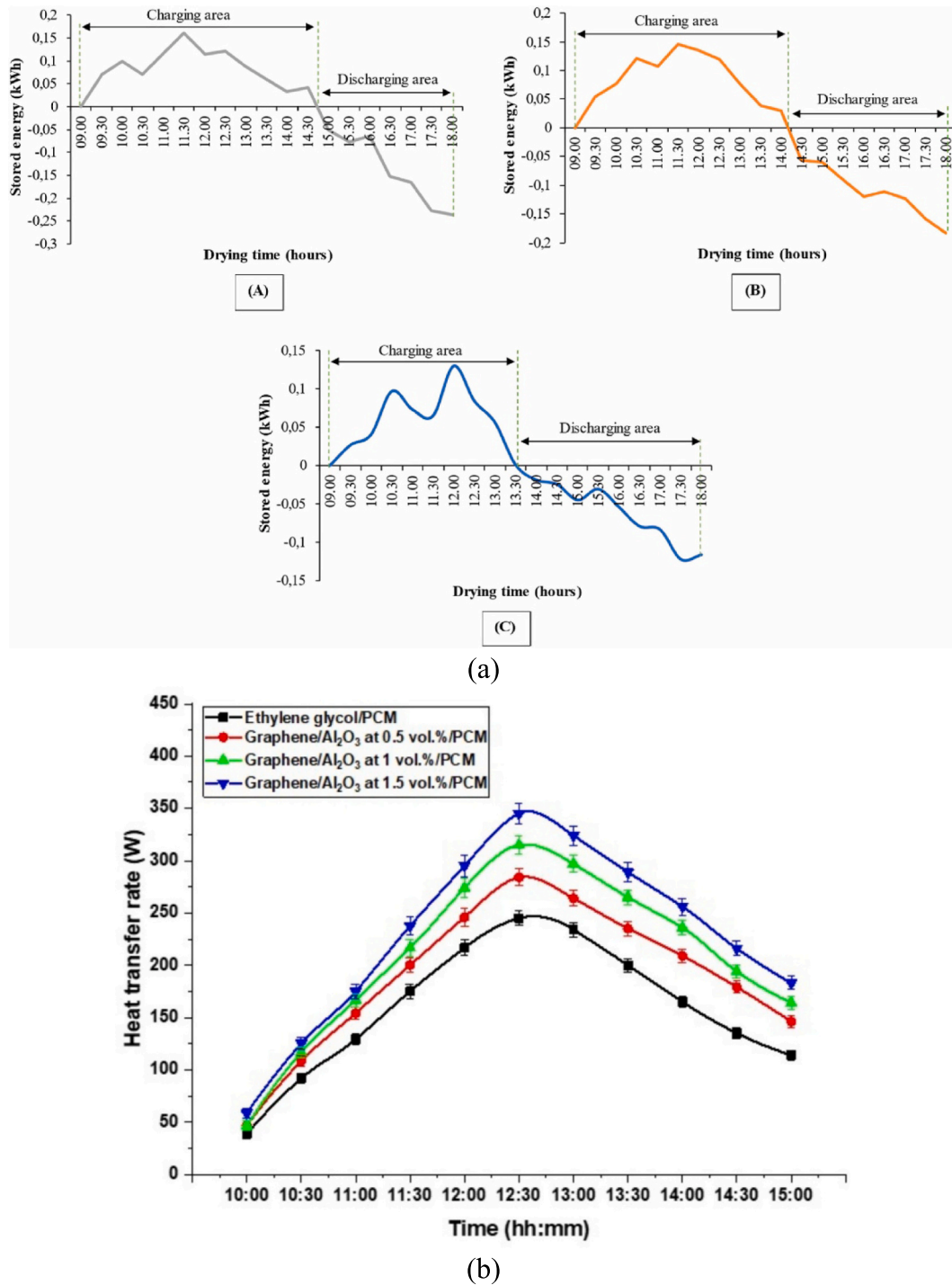


Fig. 9. (a) Charging and discharging of PCM during the drying process at air recycling ratios 100 % (A), 50 % (B), and 0 % (C) [104], (b) effect of PCM on heat transfer rate [105], (c) effect of PCM thickness on the performance of solar air collector [59], (d) effect of PCM locations temperature and liquid fraction of PCM in ISD [106], (e) temperature variation for mango pulp drying in different PCMs of solar dryer [109], and (f) charging and discharging of PCM in the solar dryer with two different masses [102].

aspects of solar dryer performance and optimization, demonstrating its utility in this niche yet important technological area. In their investigation, Lillo-Bravo [120], employed a random forest model to forecast the operational effectiveness of a solar water heating system. The predictive capabilities of the model were then evaluated based on key performance indicators. Specifically, the Mean Absolute Percentage Error (MAPE) was used as a metric, revealing a range of error between 2.94 % and 5.86 % for the predicted annual energy supplied by the

system, as well as for the predicted solar fraction. Complementing this, the coefficient of determination, or R-Square, was also assessed. The results indicated a strong correlation between the predicted and actual values, with R-squared values ranging from 0.995 to 0.998 for the annual energy supplied. This demonstrates a high degree of accuracy in predicting the overall energy output of the solar water heating system. Zhang et al. [121] used a random forest algorithm to estimate biomass MC in a fluidized bed dryer for real-time monitoring. The model

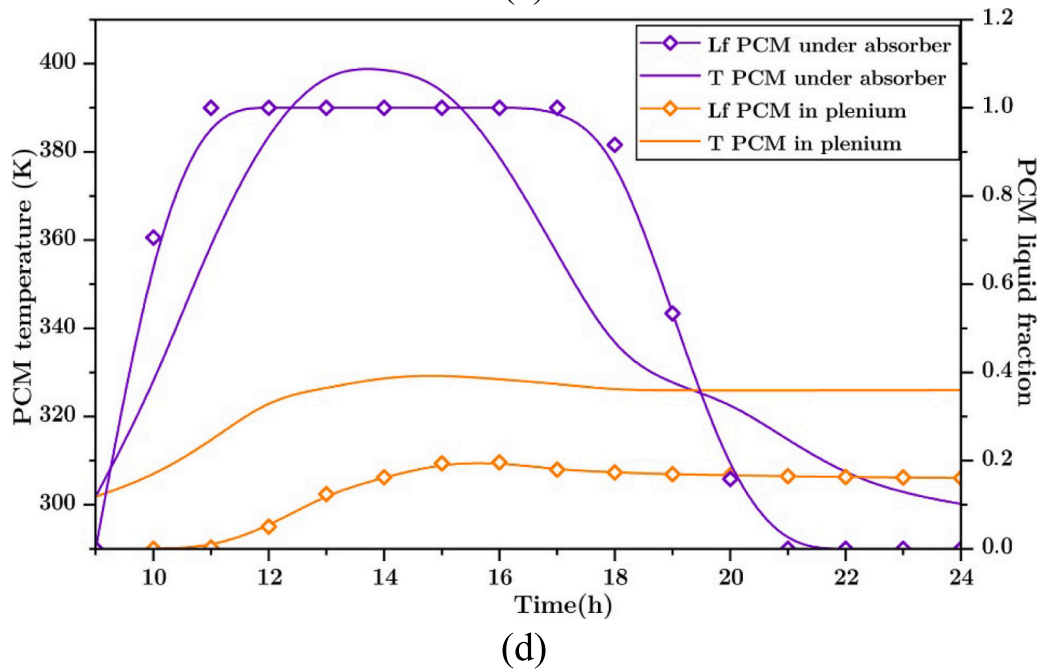
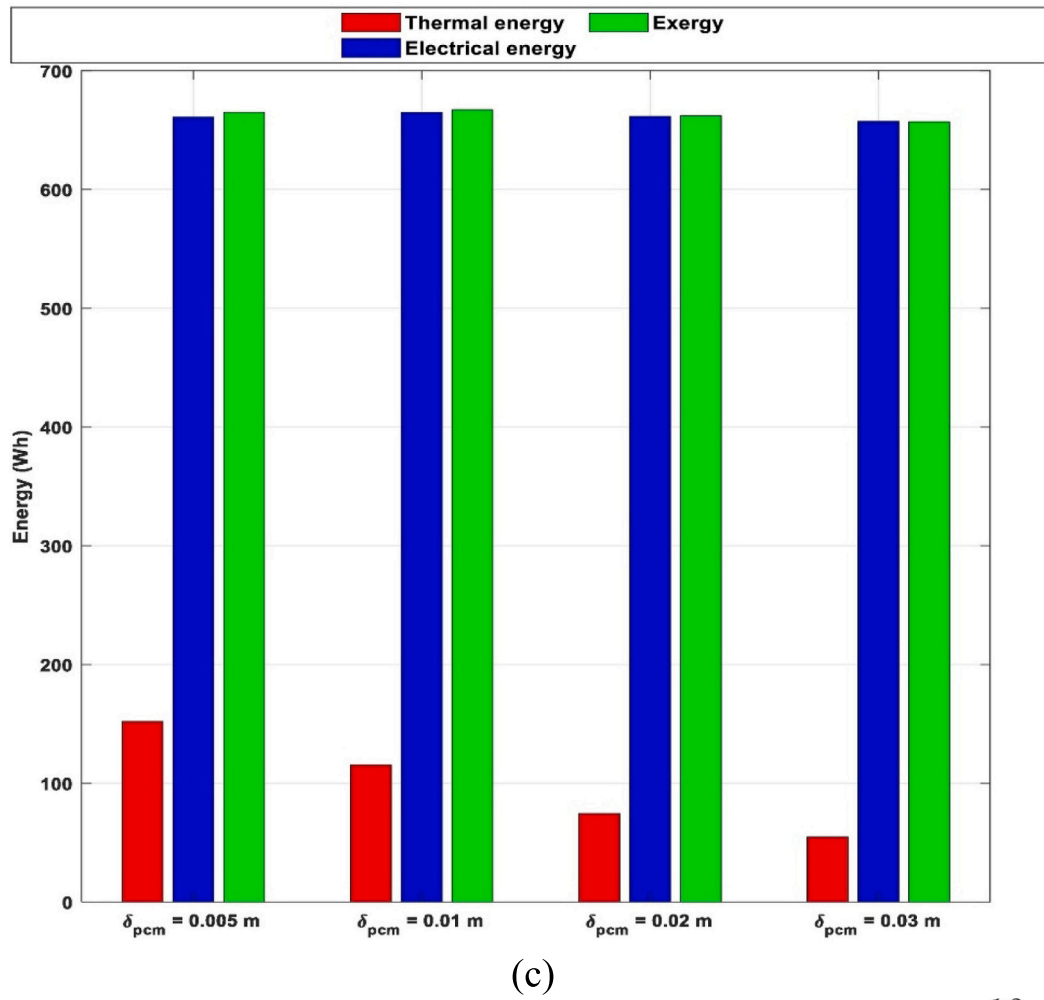


Fig. 9. (continued).

achieved promising accuracy, with a maximum relative error below 13 % compared to reference values. This suggests that the random forest model is a viable solution for online monitoring and control, potentially

improving the efficiency of fluidized bed drying. The findings demonstrate the algorithm's potential for providing accurate and timely moisture information during the drying process. Abdelkader et al. [122]

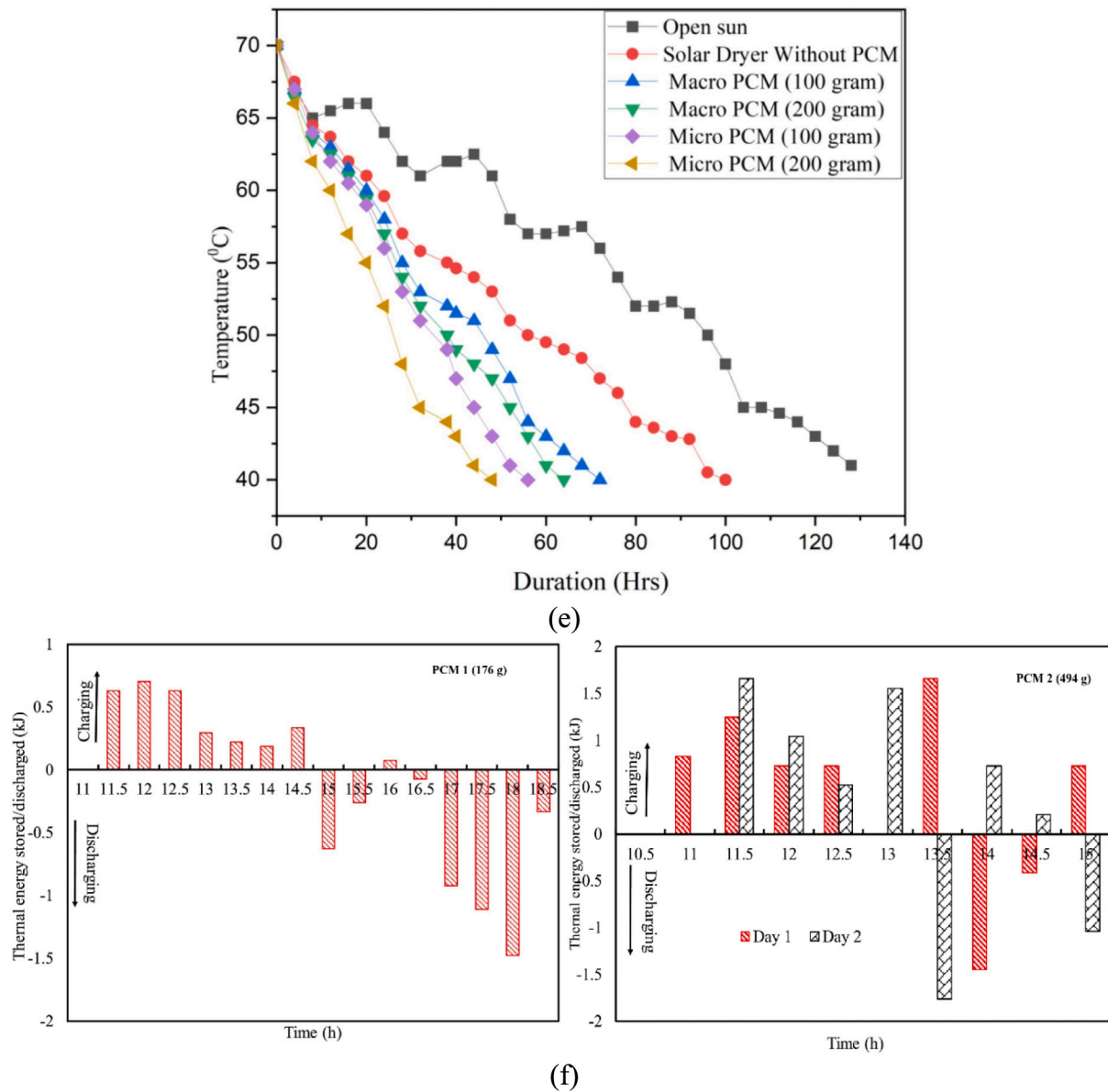


Fig. 9. (continued).

evaluated the performance of an indirect solar dryer for potatoes using a neural algorithm. Their analysis, focusing on energy and exergy efficiency and environmental impact, aimed to assess the dryer's potential for sustainable potato drying. The random forest algorithm indicated an overall drying efficiency of approximately 15 %. Chahboun and Maarofi [123] have shown that using the random forest algorithm in predicting solar energy production has better performance than other machine learning techniques, and even by adding a series of features to our input, such as solar radiation and temperature. Malashi et al. [47] investigated the microwave drying kinetics of purple carrots, using the random forest algorithm to model the process. Their results showed that the random forest method was the most effective for estimating drying kinetic parameters, supported by a high coefficient of determination ($R^2 = 0.96$) and a low MAPE = 0.03. This indicates that the random forest algorithm is a suitable and reliable tool for modeling the microwave drying of purple carrots.

5.3. Support vector machine

SVM stands out as a highly popular and powerful ML algorithm,

finding extensive application within the realm of supervised learning. Its effectiveness stems from its unique approach to model training. The SVM algorithm trains its model through a comprehensive analysis of the provided data, diligently learning patterns and relationships embedded within. Furthermore, SVM doesn't just passively observe the data; it actively and continuously improves its predictive ability over time. Through iterative adjustments and refinements based on the training data, the model becomes increasingly adept at making accurate predictions [31,124,125]. This versatile model possesses the inherent capacity to handle a diverse range of machine learning problems, specifically demonstrating the ability to perform both classification and regression tasks. Whether it's categorizing data points into distinct groups (classification) or predicting continuous values (regression), SVM provides a robust and adaptable solution. Its ability to handle both types of tasks makes it a valuable tool for a wide variety of applications [126]. Vapnik and his research team initially introduced and meticulously developed this particular algorithm, which has since achieved substantial adoption across a diverse spectrum of data science and machine learning applications [127]. Its widespread utilization can be observed in areas such as facial recognition systems, optical character

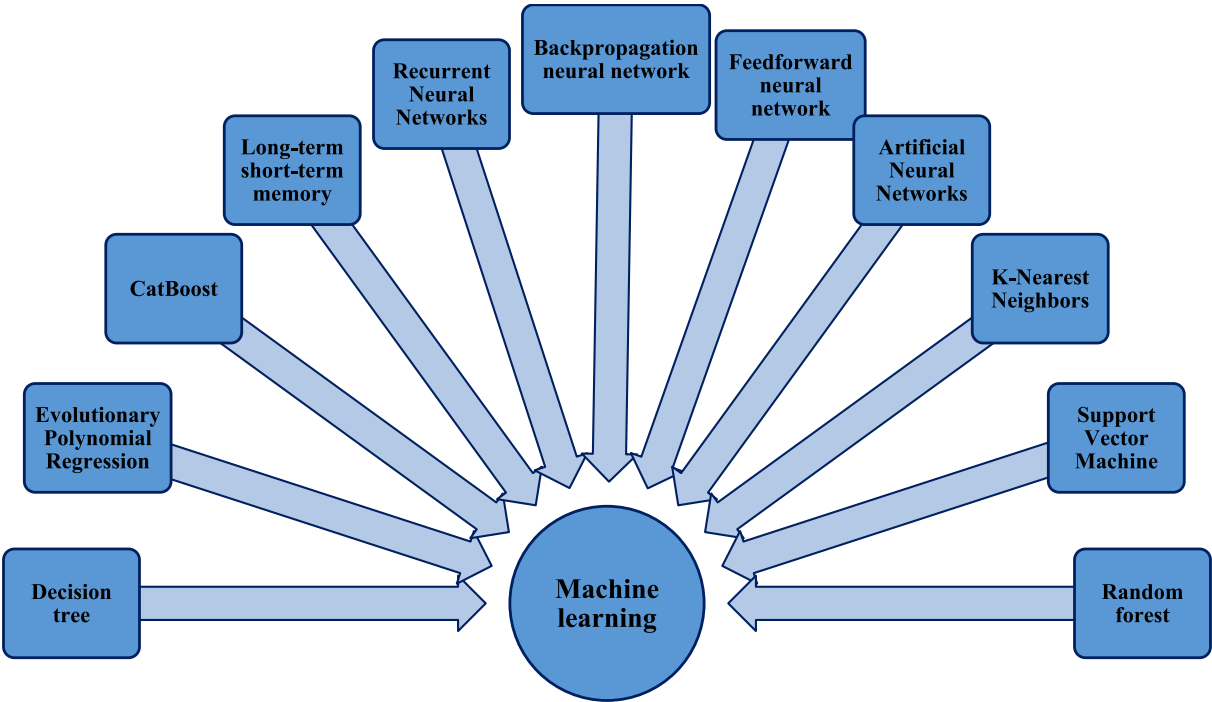


Fig. 10. Algorithms used in the drying process using a solar dryer.

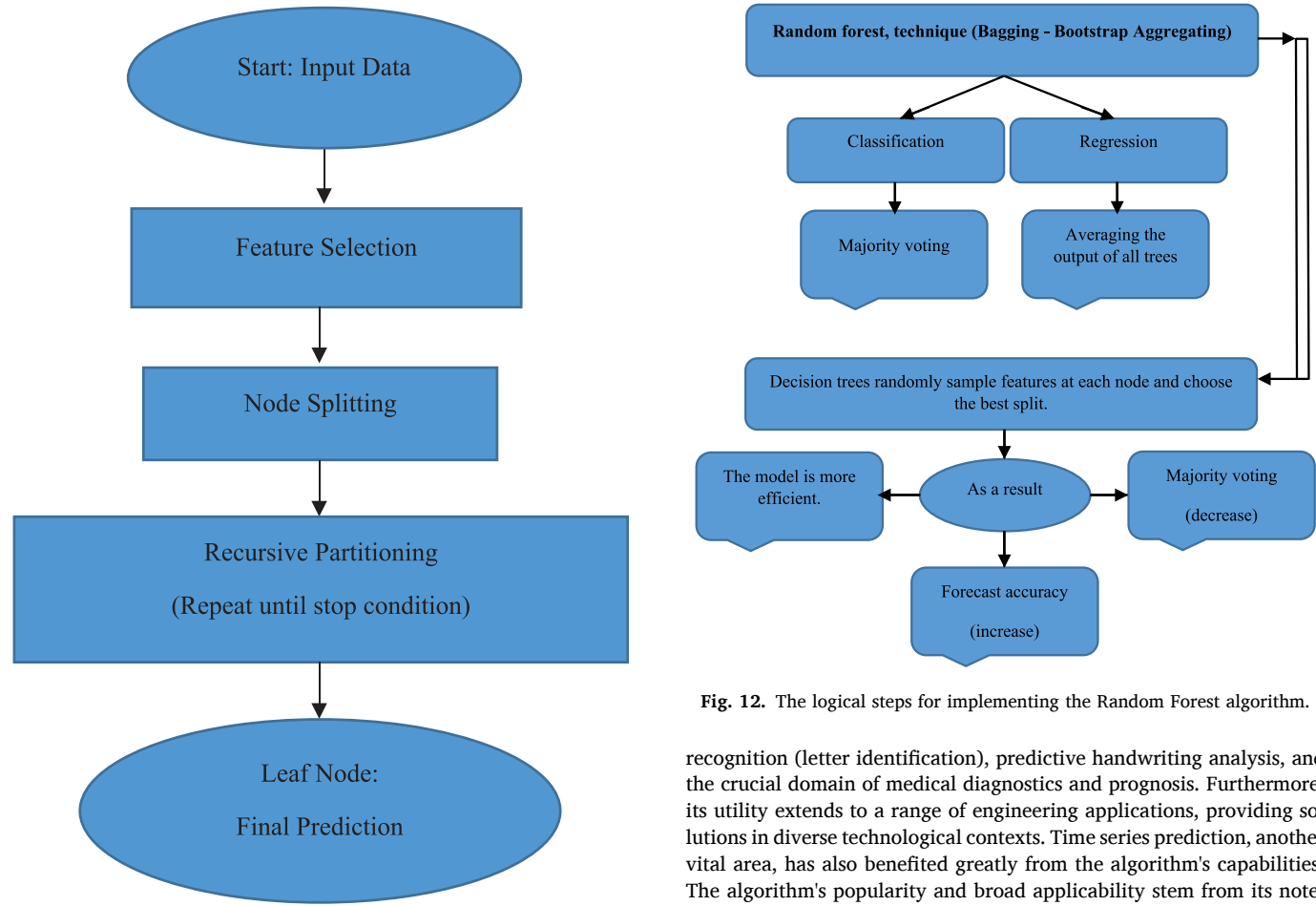


Fig. 11. Diagram of the decision tree algorithm based on its performance.

Fig. 12. The logical steps for implementing the Random Forest algorithm.

recognition (letter identification), predictive handwriting analysis, and the crucial domain of medical diagnostics and prognosis. Furthermore, its utility extends to a range of engineering applications, providing solutions in diverse technological contexts. Time series prediction, another vital area, has also benefited greatly from the algorithm's capabilities. The algorithm's popularity and broad applicability stem from its noteworthy characteristics, which include a high degree of accuracy in predictive modeling, a strong capacity for generalization to unseen data, and an inherently optimal structural design. These attributes collectively

contribute to its effectiveness and robustness, making it a preferred choice in numerous predictive modeling scenarios. The references [44,128–131] provide detailed support for these observations and further insights into the algorithm's performance and applications. Fig. 14 shows part (a) of the SVM diagram for further understanding, and part (b) shows the key steps of the SVM algorithm.

SVM finds the optimal hyperplane to separate data points into classes, maximizing the margin between them. Using kernel tricks, SVMs handle both linear and nonlinear data, making them versatile for various supervised learning tasks in energy research [132–134]. The equations related to the SVM algorithm are as follows [135]:

$$W^T + b = 0 \quad (4)$$

$$Y(x) = \frac{1}{N} \sum_{n=1}^N \omega^T k(x_i, x_j) + b \quad (5)$$

$$W^T(x) + b = 0 \quad (6)$$

$$k(x_i, x_j) = (x_i)^T (x_j) \quad (7)$$

where the parameters ω and b represent the weight and bias vectors, respectively. Also, the parameters n , λ , and ϕ represent the number of samples, the regularization factor, and the mapping function, respectively. Fig. 15 shows the main structure of the SVM algorithm.

5.4. Support vector regression

SVR is a powerful and versatile regression technique that builds upon

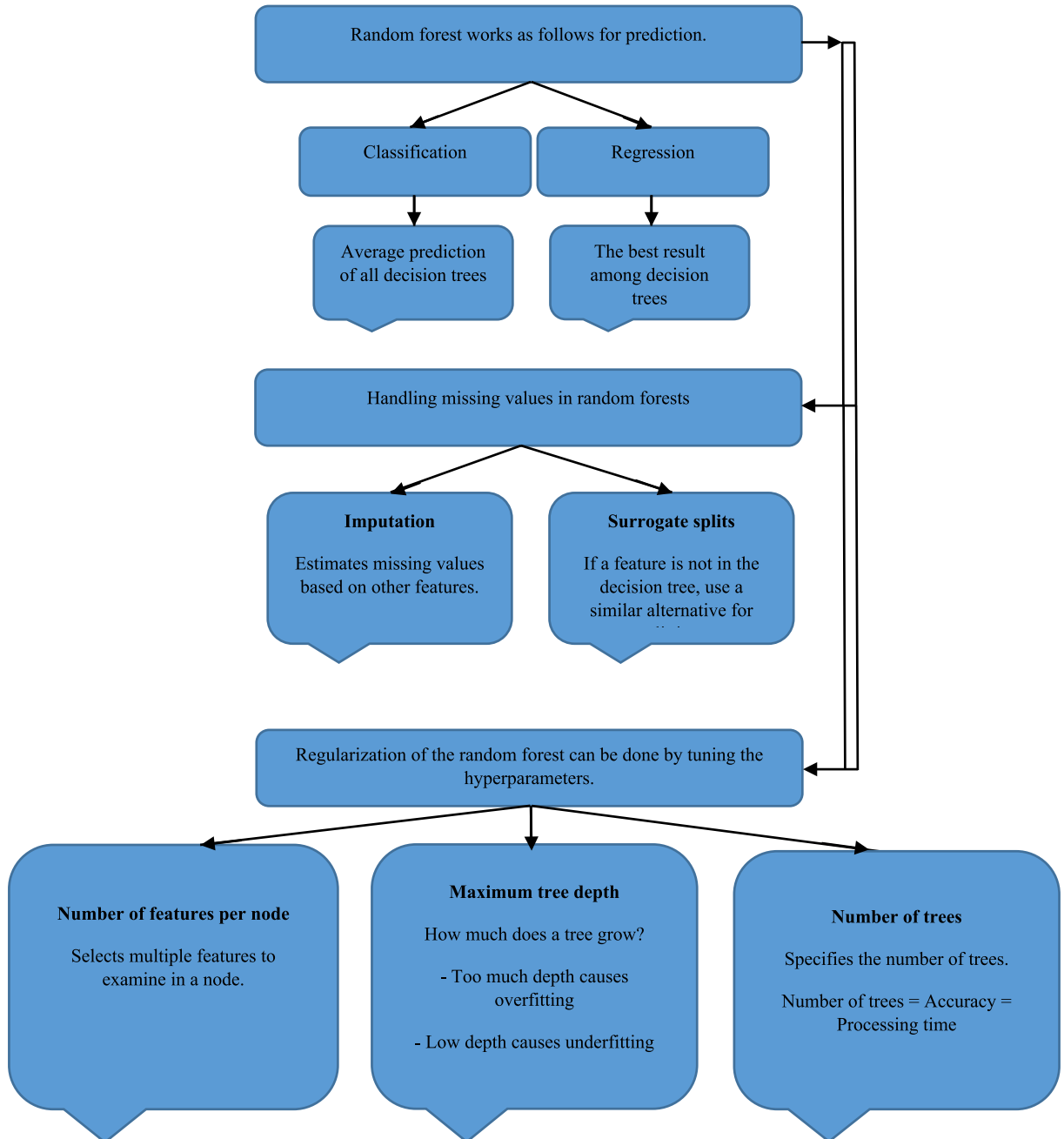


Fig. 13. The evaluation of each node of each decision tree.

the principles of the well-established SVM algorithm. Essentially, SVR can be viewed as a regression-focused extension of SVM. While both algorithms share a fundamental mathematical foundation, their intended applications and ultimate goals differ. SVR is specifically designed for regression tasks, aiming to predict continuous values, whereas SVM is primarily used for classification problems, assigning data points to distinct categories. As a key method within the realm of supervised ML, SVR is recognized for its effectiveness in building predictive models from labeled data [136,137].

One of the primary reasons for SVR strong performance lies in its unique approach to finding the optimal function. Instead of striving to minimize the error between predicted and actual values for every single data point, SVR seeks a function that accurately represents the majority of the data within a defined margin of tolerance, known as the ϵ -insensitive tube. This tube essentially creates a buffer around the predicted values, allowing for a certain degree of deviation without penalizing the model excessively. The algorithm simultaneously aims to minimize the complexity of the function itself, avoiding unnecessary

intricacies [31,138]. This dual objective – fitting the maximum amount of data within the ϵ -insensitive tube while minimizing model complexity – allows SVR to effectively balance the crucial trade-off between achieving high prediction accuracy and preventing overfitting. Overfitting, a common pitfall in machine learning, occurs when a model becomes too closely tailored to the training data, capturing noise and irrelevant patterns that hinder its ability to generalize to unseen data. Maintaining this balance is especially critical when modeling complex processes, such as solar drying, where inherent nonlinearities and fluctuating operating conditions present significant challenges [49,139]. The ability of SVR to navigate these complexities makes it a valuable tool in such domains. In environments marked by nonlinearity and variable operational settings, avoiding overfitting while maintaining predictive power is paramount. The main steps of the SVR algorithm are shown in Fig. 16, which is very important.

In recent years, researchers in the field of drying have used the SVR algorithm. For this purpose, they used an infrared radiation convection device for the drying process and predicted the MC parameter using the

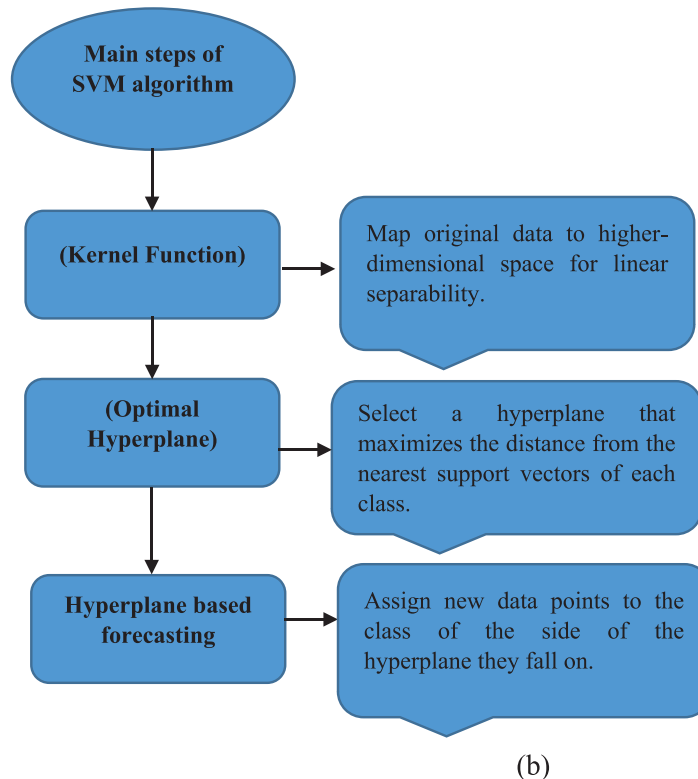
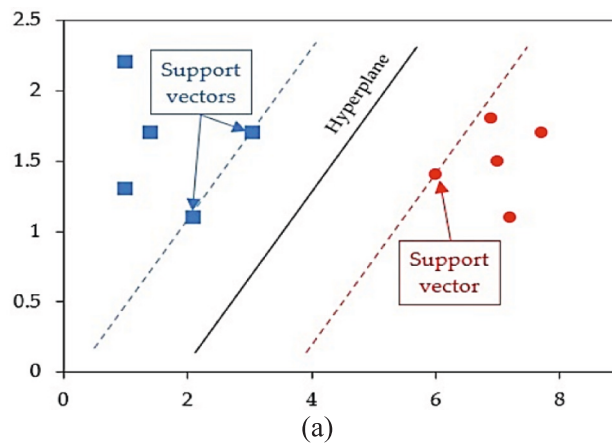


Fig. 14. (a) SVM chart [31], (b) main steps of the SVM algorithm.

SVR algorithm, which yielded appropriate values such as MSE and R^2 equal to 0.000169 and 99.8 %, respectively [140]. Also, in another study, the SVR algorithm was used to estimate the MC parameter and compare it with experimental data for drying lentil seeds using a microwave dryer, and its results showed good prediction accuracy with R^2 values of 0.995 [49]. In another recent study, tobacco drying was performed using a heat pump dryer, and its optimization was performed using the SVR method. Concluded that the calculation time was reduced and the optimal MSE reached a value of 0.021738 [141]. Yagli et al. [142] analyzed the performance of ML algorithms under varying climatic conditions, primarily for energy prediction. However, SVR is also suitable for modeling drying processes because they are significantly influenced by climate [142,143].

SVR aims to find a function, $f(x)$, that approximates target values, y_i , within a tolerance margin, ϵ (epsilon). As illustrated in Fig. 17, SVR attempts to fit a “tube” around the data, predicting values within ϵ of the actual target. This is achieved by defining the relationship between input features (x) and predicted output using a linear or kernel function. For simplicity, assume a linear relationship for the regression model is based on the following line [144]:

$$f(x) = \langle \omega, x \rangle + b \quad (8)$$

SVR aims to minimize the margin subject to the condition outlined in [145].

$$\begin{cases} y_i - \langle \omega, x_i \rangle + b < \epsilon \\ \langle \omega, x_i \rangle + b - y_i < \epsilon \end{cases} \quad (9)$$

Finally, it can be calculated from the following equation.

$$\frac{1}{2} \|\omega\|^2 \quad (10)$$

5.5. K-Nearest Neighbors

The KNN algorithm, a fundamental technique in the field of machine learning, has a history stretching back to the 1950s, where it was initially presented by several researchers and authors working independently. Since its inception, KNN has become recognized as a method that is both straightforward to understand and apply, and exceptionally popular among a range of different machine learning approaches. More specifically, KNN falls under the category of instance-based learning, meaning it relies directly on the training instances rather than learning explicit parameters. This instance-based nature makes the KNN

algorithm particularly valuable in scenarios involving pattern recognition. Its utility also extends to classification tasks, where the aim is to assign new data points to predefined categories. The breadth of KNN's applicability is further demonstrated by its use across numerous distinct applications within diverse fields. The core principle underpinning the KNN algorithm is that of proximity or similarity. When presented with a new, unclassified element, the algorithm determines its category based on the characteristics of the training examples that are nearest to it within a defined feature space. In simpler terms, each new data point is classified by examining the ‘K’ closest existing data points in the training set, as measured in the feature space, and assigning the new data point to the class most represented among these neighbors [147–149].

The KNN algorithm offers a range of benefits and applies to various tasks. Notably, KNN distinguishes itself by its non-parametric nature, meaning it does not necessitate adherence to predefined statistical distributions or assumptions about the underlying data. This flexibility is a significant advantage in scenarios where the data's characteristics are unknown or complex. Furthermore, the versatility of KNN extends to its ability to address different types of problems. Specifically, it can be effectively employed for classification tasks, where the goal is to assign data points to distinct categories. It is also suitable for regression problems, which involve predicting continuous numerical values. In addition to these, KNN can be utilized for sorting problems, effectively organizing data based on similarity or proximity. These diverse applications, spanning classification, regression, and sorting, are well-documented in the existing literature [150–152]. The main steps of the KNN algorithm are shown in Fig. 18(a).

Fig. 18(b) illustrates the KNN algorithm, which classifies a new data point based on the majority class of its K nearest neighbors. With $K = 3$, the algorithm classifies the central circle as belonging to the dotted triangles because they constitute the majority among the three nearest neighbors. However, increasing K to 11 results in a different classification: the central circle is classified as belonging to the solid triangles, which now represent the majority class within the expanded neighborhood. This demonstrates the sensitivity of KNN to the choice of K, as even though the circle is visually closer to the triangles, the algorithm prioritizes the majority class within the defined neighborhood.

Keramat-Jahromi used an electro hydrodynamic dryer to estimate MC. The results of evaluating the MC parameter with the KNN algorithm are equal to $R^2 = 0.959$, respectively [154]. The following are two significant uses of the KNN algorithm in hybrid models:

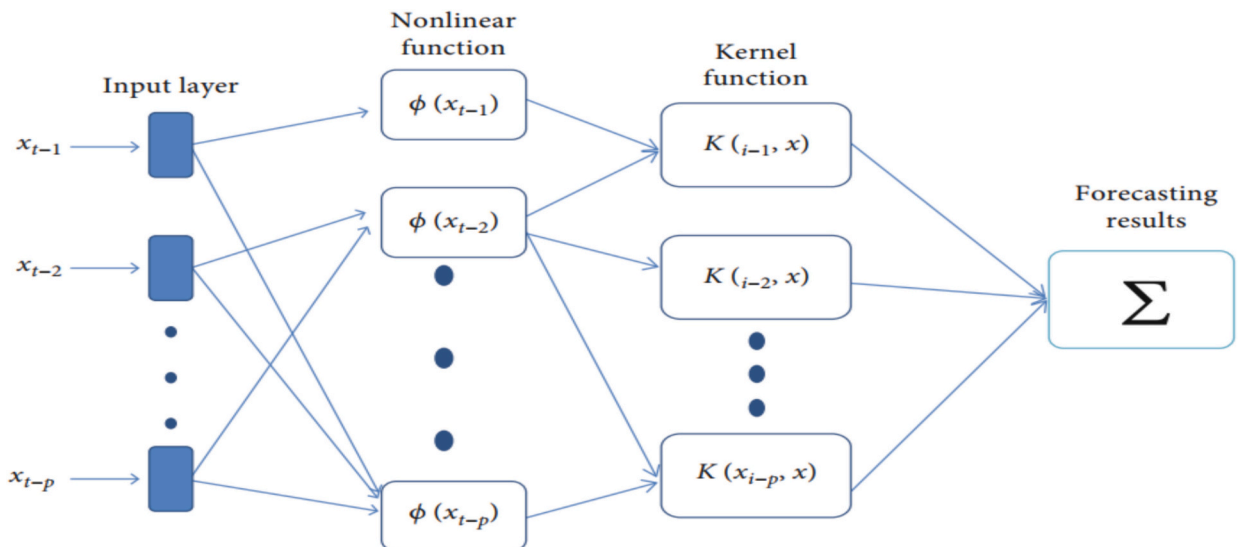


Fig. 15. General structure of the SVM algorithm [135].

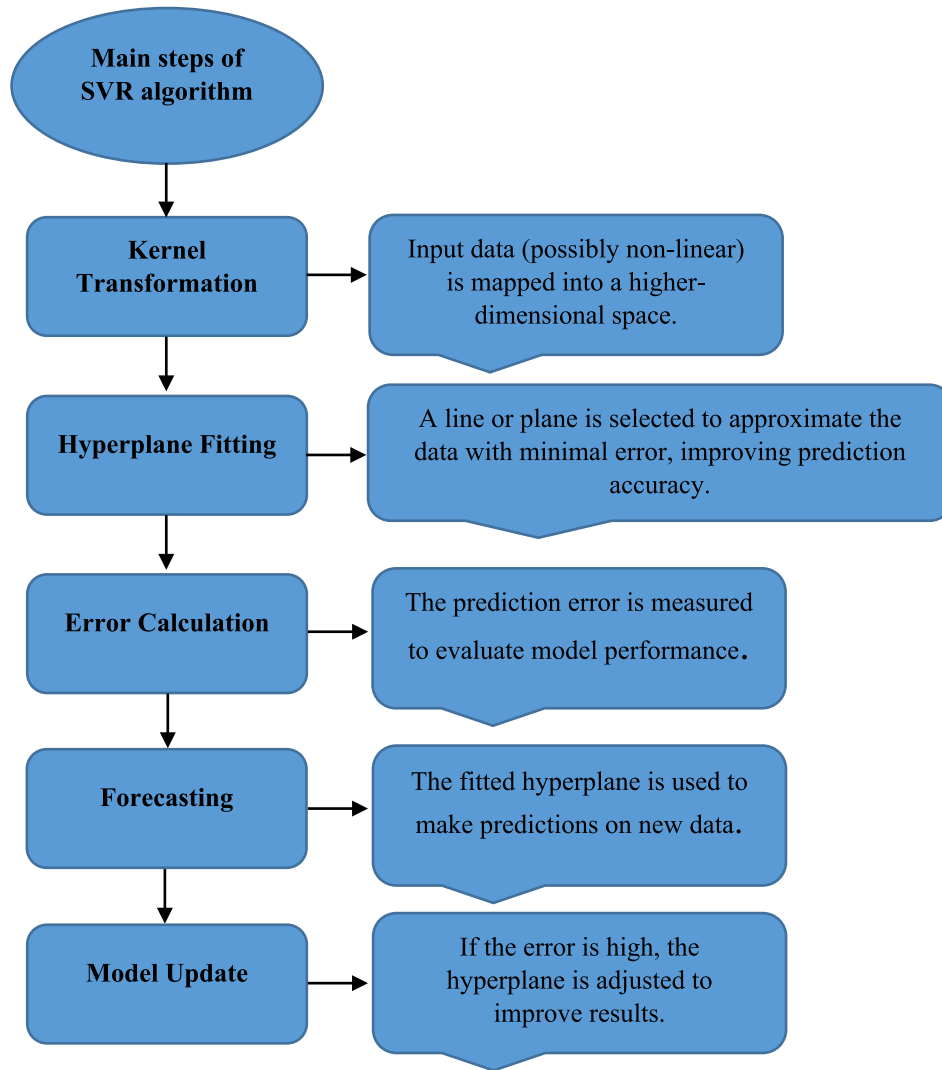


Fig. 16. Main steps of the SVR algorithm for parameter estimation.

1. ANN uses neural network-extracted features to improve prediction accuracy when used in conjunction with the KNN method [155].
2. To enhance performance in solar power forecasting, the KNN algorithm leverages structural variety and adaptability in conjunction with SVM [149].

5.6. Artificial neural networks

ANNs, inspired by the human brain's structure and parallel processing, are a powerful class of machine learning algorithms used in

diverse fields. Their bio-inspired design enables them to excel at pattern recognition and relationship modeling within large datasets, particularly complex and nonlinear relationships. Through iterative adjustments based on data exposure, ANNs learn to refine their understanding of underlying relationships. They often outperform traditional modeling methods when dealing with intricate or nonlinear data, providing more accurate and robust representations that lead to improved predictive performance [156,157].

The first steps in research in the field of artificial neural networks date back to the 1940s, when McCulloch and Pitts presented a conceptual model of the functioning of neurons [78,158]. Artificial neural networks, as flexible computational tools, have similar performance to nonlinear regression methods and are used in engineering and scientific problems due to their high generalizability, noise resistance, and ability to model highly complex and dynamic systems. This makes them effective for analyzing diverse phenomena, including drying processes. Supervised learning algorithms, especially ANNs, are crucial in the drying industry for accurately estimating process variables and simulating system dynamics. Their predictive capabilities enable optimized drying operations and ensure product quality, further enhanced by their adaptability to specific drying scenarios [159,160]. The basic types of artificial neural networks are shown in Fig. 19.

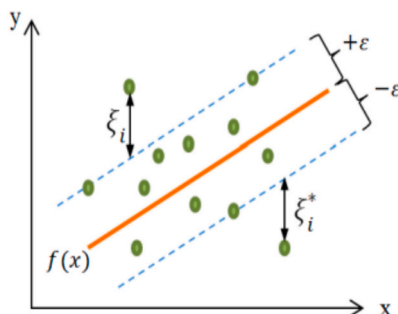


Fig. 17. SVM regression structure [146].

5.6.1. Architecture of ANN

ANNs are usually described as an arrangement of interconnected nodes organized into three distinct layers. The first layer, known as the “input layer,” consists of nodes that receive external information and feed it into the model. The second layer consists of one or more “hidden layers”; in these layers, each node receives signals from the previous layer, processes them through combination and transformation, and passes the resulting output to the next layer. The final layer, the “output layer,” consists of one or more nodes that aggregate the processed information from the last hidden layer and produce the final output of the model [161,162]. Most ANN models use a multilayer architecture that includes one or more hidden layers. The design and configuration of these models directly depend on the nature of the problem being studied. Key parameters, such as the choice of input variables, the number of hidden layers, the number of nodes in each hidden layer, and the type of transfer (activation) functions, must be chosen carefully (Fig. 20). These parameters are usually determined through an iterative process of trial and error to identify the most effective configuration for the intended application. This inherent flexibility in the architecture is one of the factors that has led to the widespread use of artificial neural networks in a variety of domains and problems [163,164].

ANNs are increasingly used in drying process research, including freeze-drying. For example, Menlik et al. employed an ANN to model the freeze-drying of apples, focusing on the relationship between process parameters and apple quality. Specifically, they used the ANN to predict MC, Moisture Rate (MR), and Drying Rate (DR). Their results, depicted in Fig. 21(a), demonstrate the algorithm's accurate predictions with low error rates, indicating strong agreement between predicted and observed values. Fig. 21(b) illustrates the input parameters used by the ANN to predict these key drying parameters [166].

Karaağaç et al. studied a microwave solar dryer system with a thermal collector, focusing on estimating MC using ANN and SVM. Comparing the algorithms' performance against experimental data, their

results (Fig. 22) showed that ANN more accurately predicted MC than SVM. Table 10 provides a complete overview of the ANN algorithm in solar dryers and its important results.

ANNs, often modeled as Multi-Layer Perceptron (MLPs), consist of interconnected neurons arranged in layers: an input layer, one or more hidden layers, and an output layer [10,174]. Weighted connections between neurons determine the influence between them. Training an ANN, especially an MLP, involves iteratively adjusting these connection weights based on input data to produce desired outputs. This adjustment, guided by the difference between the network's output and the target, allows the network to learn complex relationships [174]. Eqs. (11) and (12) rigorously define the output functions of individual neurons and the overall network error (E_r), providing a framework for optimization.

$$q_i = f \left(\sum_{j=1}^{L_M} (W_{ij} \cdot P_i) \right) \quad (11)$$

$$E_r = \frac{1}{N} \sum_{i=1}^N (E_i - q_i)^2 \quad (12)$$

Given input data P_i and weight factors W_{ij} , an activation function $f(\dots)$ computes the net output q_i . This output is then compared to the expected output E_i .

5.7. Feedforward neural network

Feed-forward Neural Networks (FNNs) are successfully applied in diverse fields due to their ability to learn from examples, obviating the need for explicit programming, and to generalize to unseen data. This makes them particularly useful for complex, nonlinear problems like predicting drying and other industrial processes. However, developing these models presents challenges, notably the absence of performance guarantees. Therefore, careful selection of training data and model settings is critical for successful application [175,176]. Fig. 23 shows the main steps of the FNN algorithm.

Feedforward controllers leveraging artificial neural networks offer a swift response to fluctuations in solar energy, primarily due to their inherent ability to forecast and adjust to variations in the surrounding environment. This predictive and adaptive capability is a key characteristic that sets them apart. In essence, these controllers can anticipate shifts in solar irradiance and proactively adjust system parameters, leading to enhanced performance. This particular attribute renders them highly suitable for a range of applications, notably in the optimization of solar energy system performance. The benefit is particularly pronounced in scenarios characterized by rapid and unpredictable environmental changes. For instance, when cloud cover fluctuates dramatically or when there are sudden shifts in temperature or atmospheric conditions, a neural network-based feedforward controller can maintain optimal operation. The ability to quickly adapt to these dynamic conditions makes them more effective compared to traditional control methods. These advantages allow solar energy systems to achieve greater efficiency and stability, particularly when operating under volatile environmental conditions [177–179].

Between the input and output layers are one or more layers that make up FNNs. Full connectivity is a property of these layered structures, meaning that every neuron in one layer is connected to every other neuron in the layer above. Fig. 24 depicts the overall layout of this architecture.

5.8. Backpropagation neural network

Back-Propagation Neural Network (BPNN) encompass a diverse set of multilayer, interconnected artificial neural networks. Employing a supervised learning algorithm rooted in gradient descent, these networks iteratively refine their weights to minimize prediction error. Their

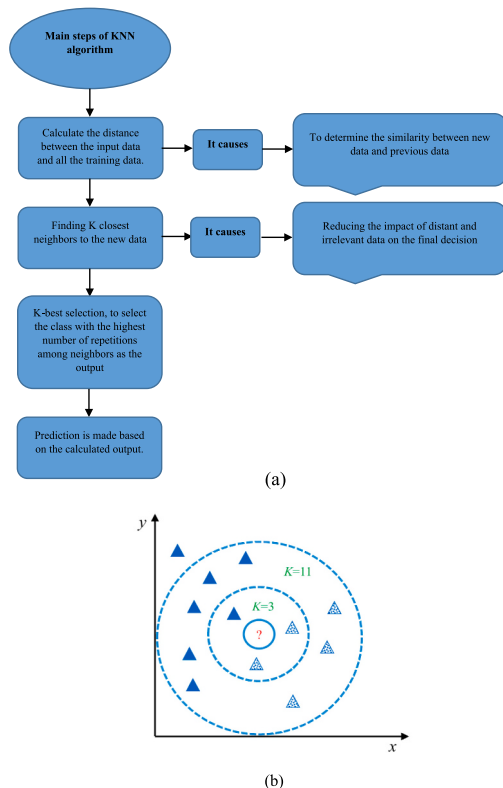


Fig. 18. (a) Main steps of the KNN algorithm in parameter estimation, (b) how to implement the K-Nearest Neighborhood method in the KNN algorithm technique [153].

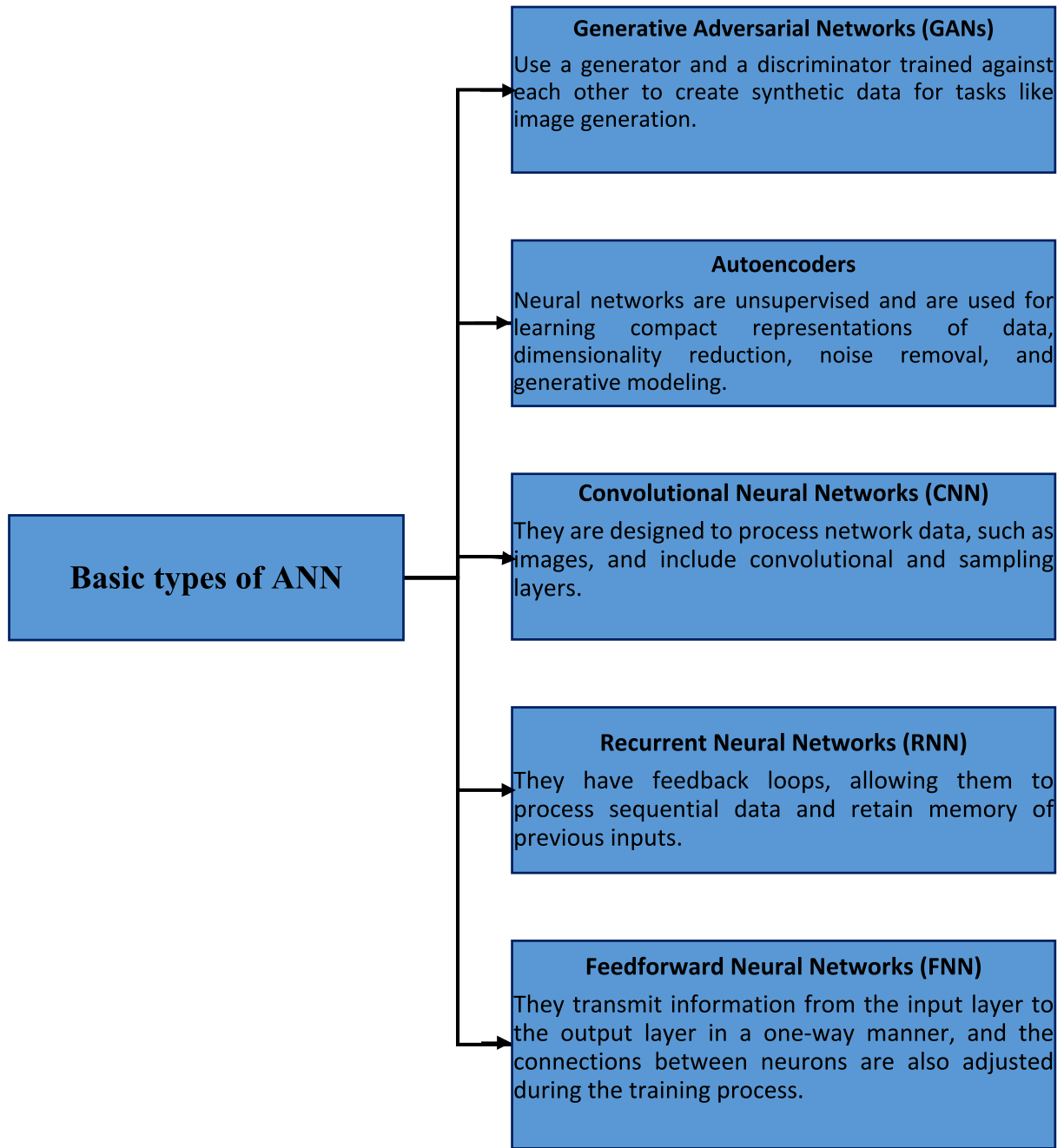


Fig. 19. Basic types of ANN.

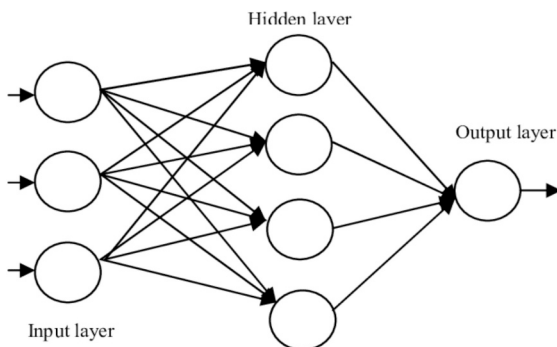


Fig. 20. Artificial neuron and multi-layered neural network architecture [165].

strength lies in modeling intricate nonlinear functions, a capability enhanced by an adequate number of hidden units. This makes them well-suited for learning complex patterns and achieving low modeling error, proving valuable for various prediction and optimization tasks across engineering and scientific domains [181,182]. The back-propagation artificial neural network algorithm consists of five main steps, which are presented in Fig. 25.

The general structure of the BPNN algorithm is shown in Fig. 26. In the following, we give the formulas for the error function and the weight function for the input and output layers, and Eqs. (13) to (17) describe the details of these weight calculations.

Eq. (13) shows the error function and the calculation of the weights.

$$e = \frac{1}{2} \sum_{i=1}^n (\bar{y} - y_i)^2, \quad \omega = \omega - \eta \frac{\partial e}{\partial \omega} \quad (13)$$

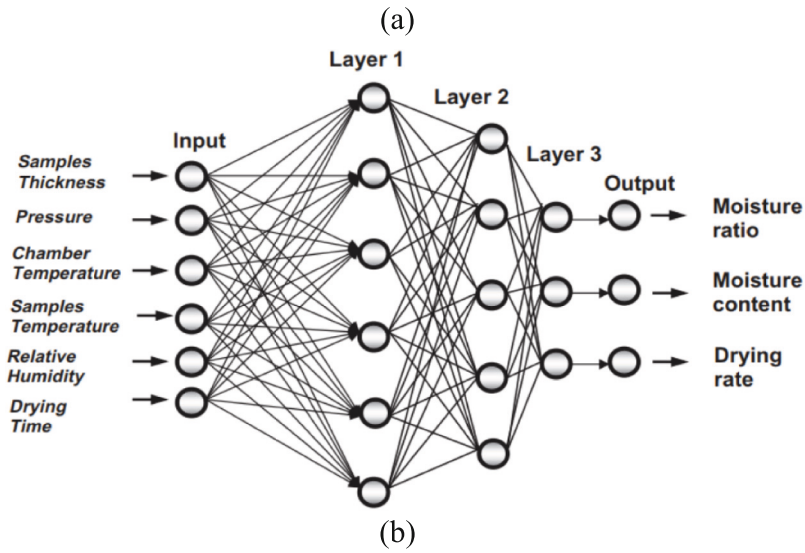
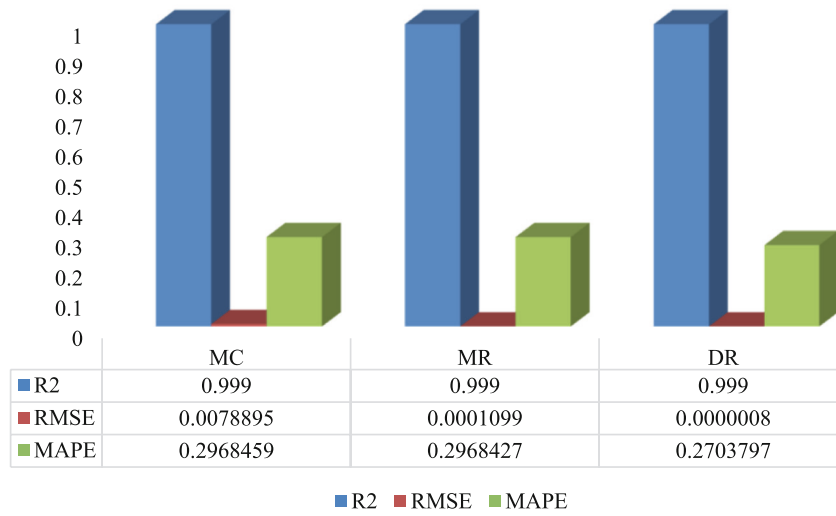


Fig. 21. (a) Estimation of drying parameters, and obtaining error, (b) ANN structure input and output parameters [166].

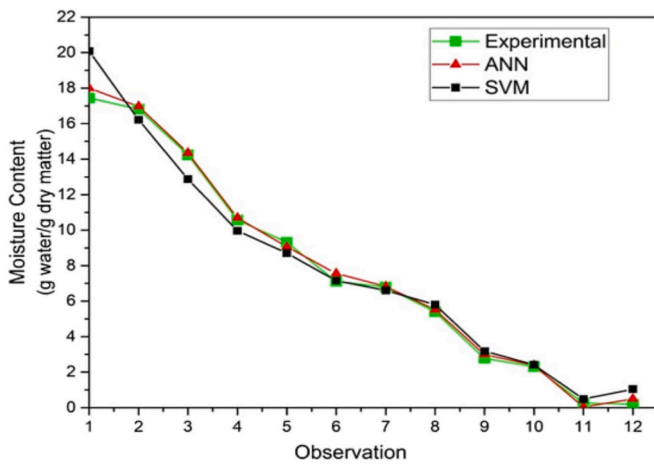


Fig. 22. Estimation of MC parameter with ANN and SVM algorithms and comparison with experimental data [44].

Eqs. (14) and (15) show the calculation of the output layer weight.

$$\omega_i^o = \omega_i^o + \eta \delta^o a_i \quad (14)$$

$$\delta^o = -(\bar{y} - y_i) f'(Z^o) \quad (15)$$

Eqs. (16) and (17) show the calculation of the hidden layer weight.

$$\omega_i^h = \omega_i^h + \eta \delta^h x_i \quad (16)$$

$$\delta^h = - \left(\sum_{o=1}^m \delta^o \omega^o \right) f'(Z_h) \quad (17)$$

5.9. Recurrent neural networks

RNNs are a vital deep learning architecture specifically designed for processing sequential data, which inherently possesses an order or temporal structure. Unlike feed-forward networks, RNNs incorporate a loop structure that acts as a memory, maintaining an internal state representing past inputs. This allows them to leverage prior information

Table 10

A review of recent research on the use of the ANN algorithm in solar dryers.

References	Dryer type	Algorithm input	Algorithm output	Key findings
[44]	concentrated photovoltaic-thermal solar dryer	$D_t, H_{dc},$ Concentrated $I_g,$ Non – concentrated $I_g,$ T_{dryer}	M_C, M_R	The estimation of the M_C parameter with ANN and SVM algorithms yields R^2 values of 0.999 and 0.974, respectively. The ANN algorithm estimated the M_C parameter well, the R^2 value reached 0.99996, and the solar air heater outlet temperature reached 45 °C
[167]	Indirect solar dryer	T_{dryer}, D_t	M_R	Using ANN, the M_C estimate for litchi flesh reached $R^2 = 0.99$ and was very accurate. The ANN algorithm performed well in estimating the drying chamber temperature (T_{cd}), and the MSE value was obtained = 0.9725.
[168]	greenhouse dryer	$t, I_g, T_{dryer}, \dot{m}$	M_C	The ANN algorithm had better results compared to mathematical drying models and had a higher correlation coefficient of $R^2 = 0.9999$.
[169]	Indirect solar dryer	$I_g, V_a, T_a, T_t,$ N_{tray}, D_{tray}	T_{cd}	ANN was able to predict the parameters $M_r, M_c,$ and D_r well with R^2 values of = 0.999, which indicates the high accuracy of this algorithm.
[170]	Pulsed vacuum osmotic dehydration	T_a, V_a, H_a, I_g, D_t	M_R	The ANN algorithm performed well in estimating total energy consumption and M_C compared to the mathematical drying model, with $R^2 = 0.996$, RMSE = 0.0002053, and MAPE = 0.4161673.
[166]	Freeze drying	T_s, P, T_{cd}, R_H, D_t	M_R, M_C, D_r	For predicting the MR of peeled apple samples
[78]	heat pump dryer	T_{dryer}, R_H, V_a	T_{ec}, M_C	
[171]	Microwave dryer	T_a, D_t	M_R	

Table 10 (continued)

References	Dryer type	Algorithm input	Algorithm output	Key findings
[172]	Hot air drying	T_s, T_a	D_K	by ANN, results were obtained with good accuracy, such that $R^2 = 0.9991$ was obtained. For predicting drying kinetics, the ANN algorithm performed well with accuracy, with R^2 and MSE values of 0.99998 and 0.000001, respectively.
[173]	Ultrasonic dryer	T_a, V_a, H_a, D_t	M_C	The ANN algorithm performed well in estimating the M_C of rice, with an R^2 value of 0.98.

during subsequent processing, enabling them to learn and adapt based on historical data interactions. This temporal dependency, where past inputs influence current outputs, makes RNNs highly effective in applications like natural language processing (where word order matters), speech recognition (where phoneme sequences are crucial), time series prediction, and dynamic process modeling. Their ability to handle sequential data and leverage temporal dependencies makes them a cornerstone of modern deep learning [184,185]. Fig. 27 shows the main steps of the RNNs algorithm.

RNNs excel at modeling temporal dependencies because they retain information from previous inputs via internal feedback loops, effectively providing short-term memory. This allows them to understand the context of current data based on past events, making them suitable for processes where order and timing are critical, such as material drying [186,187]. RNNs intricate structure enables analysis of complex time sequences, including meteorological data (temperature, pressure, wind) and renewable energy data, where temporal relationships are crucial [188]. For example, RNNs demonstrate superior performance in solar radiation forecasting compared to traditional ANNs, leading to higher accuracy and reduced forecast errors [186]. The general structure of the RNNs algorithm is shown in Fig. 28.

5.10. Long-term short-term memory

A sophisticated recurrent RNN architecture created especially to handle long-term dependencies in sequential data is represented by long-term short-term memory (LSTM) networks. LSTM was created to address major drawbacks of conventional RNNs, namely the vanishing gradient issue, and was first presented by Hochreiter and Schmidhuber in 1997 [31]. LSTM networks utilize memory cells that are capable of retaining information over long periods, enabling efficient learning of long-term temporal patterns. This capability is particularly valuable in modeling complex processes, such as food drying, where long-term dependencies play a critical role. The input, forget, and output gates in the architecture control how information moves across the network. For the model to capture dynamic, time-evolving patterns, these gates enable the selective retention or discarding of input [186,190]. The LSTM algorithm's primary steps are displayed in Fig. 29.

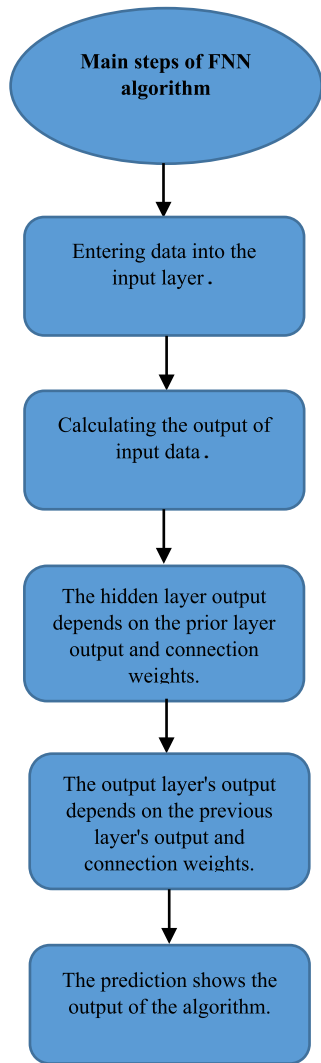


Fig. 23. The main steps of the FNN algorithm.

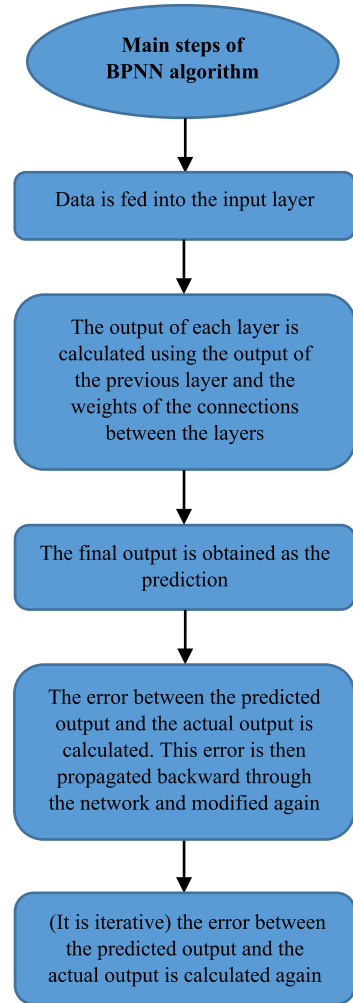


Fig. 25. The main steps of the backpropagation artificial neural network algorithm.

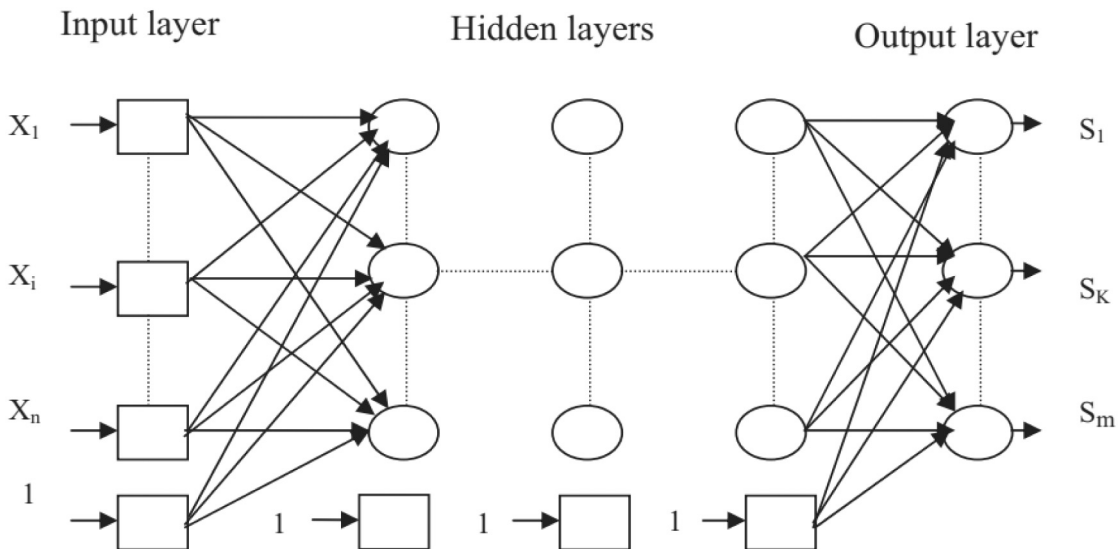


Fig. 24. The general structure of the FNN algorithm [180].

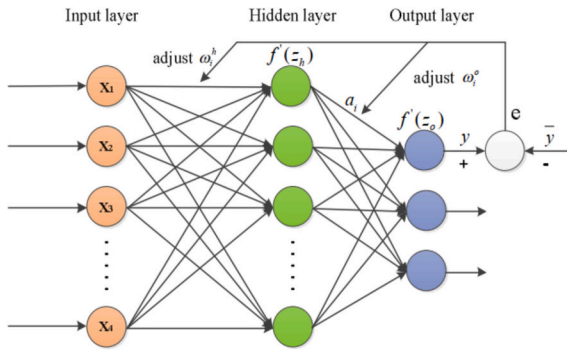


Fig. 26. General structure of the BPNN algorithm [183].

5.11. CatBoost

CatBoost, developed by Yandex in 2017, is a gradient boosting algorithm notable for its robust handling of categorical features. It addresses overfitting by sorting training data based on categorical feature values before constructing decision trees, leading to more accurate and reliable predictions [191]. CatBoost utilizes memory-efficient one-hot sequential encoding for batch features, reducing computational costs and improving performance. This approach is key to CatBoost's efficient handling of batch features [31]. For classification tasks with many classes, this algorithm uses the Target Statistic method to reduce variance and improve prediction accuracy, as demonstrated in previous research [31]. This flowchart (Fig. 30) provides a simplified representation of the main process behind the CatBoost algorithm. The process begins with input data and an initial decision based on whether a particular feature exceeds a certain threshold, which reflects the symmetric tree structure used in CatBoost. Based on this threshold, the data is split (T_1/T_2), and each path leads to outputs and initial predictions. These initial predictions are then used to compute residuals, followed by the calculation of target statistics for the batch features in a permutation-free, leak-free manner (Stat). These statistics are subsequently updated (Upd) to refine the model and complete a regular boosting iteration before the process ends (End).

CatBoost efficiently handles large datasets through parallel processing and GPU utilization, accelerating computations for data-intensive tasks. This makes it well-suited for applications like solar power forecasting and material drying modeling, where analyzing extensive data is crucial for accurate predictions and understanding complex relationships [192,193]. In an experimental study on the drying process of red beetroot, CatBoost performed more accurately than other machine learning models such as XGBoost and LightGBM. The model was able to provide better results based on metrics such as R^2 , MSE, and MAE [191].

5.12. Evolutionary polynomial regression

Grey-box models represent a middle ground between white-box models, which are based on complete knowledge of the system's internal workings, and black-box models, which treat the system as a "black box" without any understanding of its internal mechanisms. Examples of grey-box models include EPR. These models can be considered conceptual because they aim to simplify the complex underlying processes of a system in order to develop mathematical equations that effectively capture the most essential aspects of the system's overall behavior. The development of these models necessitates a simplification of the intricate processes at play within the system. One crucial step in developing grey-box models is parameter prediction. This step involves calculating the values of input and output data within a specific, pre-defined parameter range. The methodology for this calculation is detailed in the work of Giustolisi and Savic [203], which provides a comprehensive

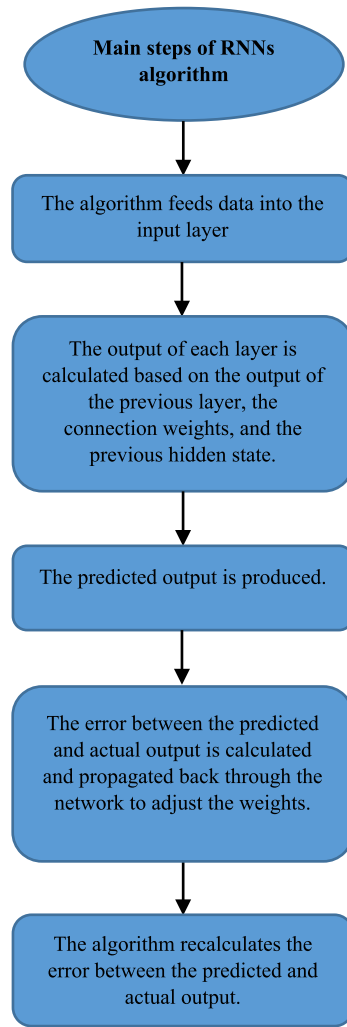


Fig. 27. The main steps of the recurrent neural network algorithm.

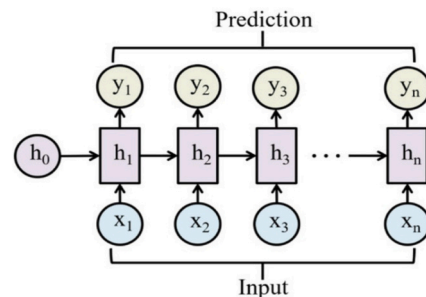


Fig. 28. The general structure of the RNNs algorithm [189].

explanation of the techniques involved in determining the appropriate parameter ranges and calculating the corresponding input and output data values for accurate parameter prediction. The calculation of input and output data within this range is essential for the proper functioning and accuracy of the grey-box model [203].

EPR has been applied to optimize solar drying systems for medicinal plants, demonstrating its utility in agricultural and environmental engineering, especially in renewable energy [7]. It uses a two-stage approach: a genetic algorithm identifies optimal model structures, and then linear regression quantifies model parameters via least squares optimization for accurate data fitting. This process yields a

comprehensive mathematical equation for prediction and analysis [7].

$$y = \sum_{j=1}^m F(X, F(X), a_j) + a_0 \quad (18)$$

This model predicts y from input data X using a function $F(X)$ and adjustable parameters a_j . An optional bias, a_0 , can be included to improve accuracy. EPR constructs equations relating X , a_j , $F(X)$, a_0 (if used), and y by selecting significant contributing factors, enabling flexible system behavior representation.

5.13. EPR vs. other intelligent optimization methods

Evolutionary polynomial regression takes a different approach compared to common machine learning algorithms and neural networks—such as artificial neural networks, support vector machines, and long-short-term memory. Models such as ANNs, SVMs, and LSTMs primarily operate as data-driven black-box systems that can identify complex, nonlinear relationships; in contrast, EPR, as a grey-box model, employs a combination of symbolic regression and genetic algorithms to ultimately generate clear and understandable mathematical equations. In terms of prediction accuracy, EPR has shown competitive performance in the solar dryer domain. For example, in estimating collector outlet temperature and thermal efficiency, the method has achieved R^2 values higher than 0.98, even outperforming results from CFD simulations (with $R^2 > 0.94$) and in some cases outperforming artificial neural network predictions (with R^2 up to 0.99) [9,48]. However, ANN and LSTM, thanks to their deep structure and ability to process time sequences, are usually more robust when dealing with high-dimensional data, time series, or highly nonlinear systems, and may provide higher accuracy in dynamic or multivariate situations, such as predicting humidity with an RMSE error of less than 0.645.

By considering computational fluid dynamic analysis speed and efficiency, EPR has a significant advantage. Unlike artificial neural networks and LSTMs, which typically require large amounts of data, fine-tuning of parameters, and long training times, EPR uses a dual evolutionary process and least-squares optimization to reach a solution faster and with less risk of overfitting. This feature makes EPR a desirable choice for situations where experimental data are limited or where

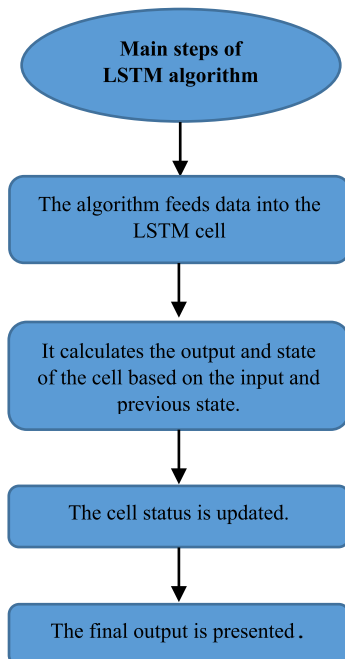


Fig. 29. The main steps of the long-term short-term memory algorithm.

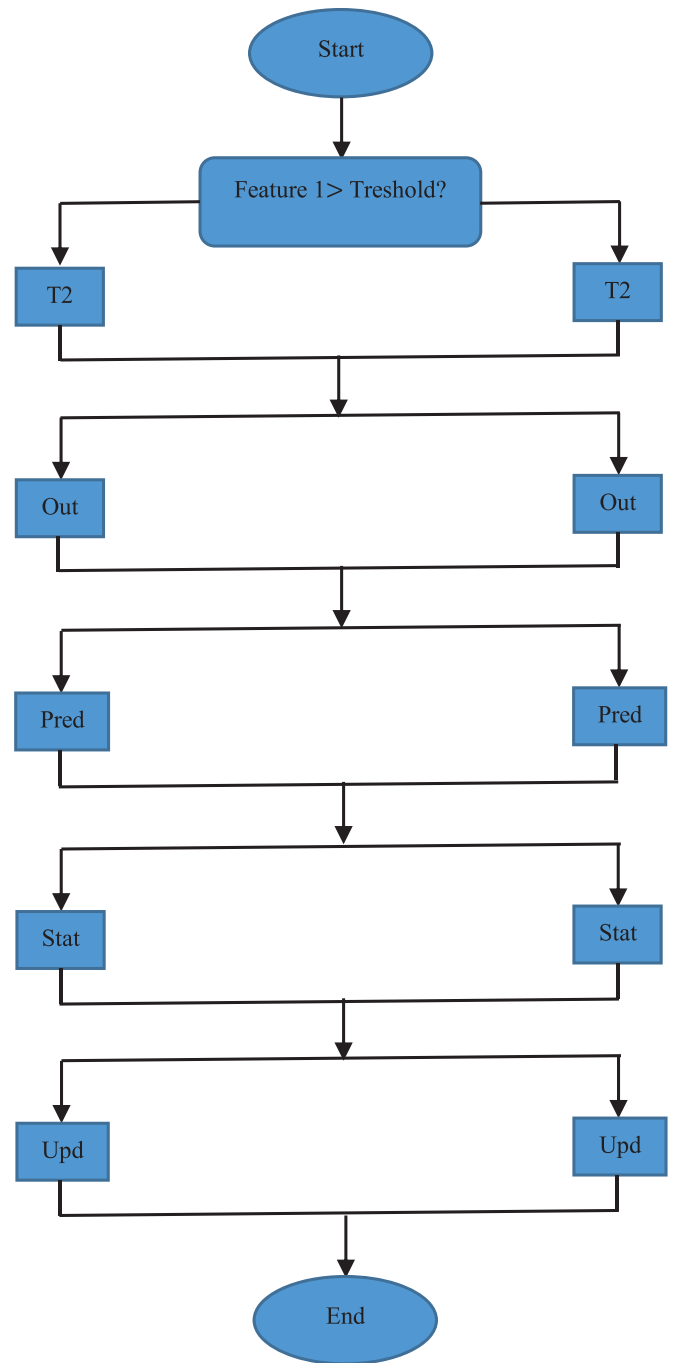


Fig. 30. Flowchart CatBoost algorithm.

model interpretability and physical explanation of the phenomenon are a priority. In conclusion, EPR is a good choice for cases where equation-based models with moderate complexity and a high level of transparency are required, while ANNs, SVMs, and LSTMs are more suitable for highly dynamic and nonlinear systems where prediction accuracy is of primary importance. Therefore, the choice of method depends on the specific needs of each project and solar drying system whether the goal is model interpretability, computational efficiency is desired, or maximum prediction accuracy is required.

6. Role of ANNs and EPR in solar dryers with PCM

Today, ANNs are used in many engineering disciplines to optimize, simulate, and model relationships between phenomena. Neural

networks were first used in 1943 by Pitt & Cilloch. These networks consist of operational elements that operate in parallel. A neural network is a simulation of the nervous system and is actually an imitation of the human brain and nervous system. In this network, an attempt is made to provide a structure that, like the brain, has the power to learn, generalize, and make decisions. Its advantages include the short time required for modeling compared to classical methods, the lack of the need for complex mathematical calculations to predict the behavior of the parameters involved in the process, and the ability to predict a limited number of experiments [177,178]. This method has become popular in all fields of science and engineering. The limitation of using this method is related to the amount of data collected from the tests, which is required for proper performance. More data is needed to train the model. Optimization of solar systems is one of the most important issues that has received the most attention of researchers in the field of solar energy today. In other words, due to the low efficiency of solar systems, there is little desire to use this type of energy systems. Therefore, many efforts are being made to increase the efficiency of the systems with various optimization methods [40,194]. A lot of research has been done in this field, which can be mentioned as follows:

In a study to predict the performance of a solar collector with PCM, soft computing method was used. For this purpose, three types of computing tools were considered, including adaptive neuro-fuzzy inference system (ANFIS), SVM, and ANN. The results showed that the ANN method with the 1-7-5 neural model with the LM learning algorithm with satisfactory R^2 values in the range of 0.832 to 0.899 can be used to predict the system performance [195]. In a project conducted by [196], in order to survey the performance of the solar dehydration system (Fig. 31), CFD and ANN was utilized. For the designed system the drying efficiency was obtained about 21.11 % to 25.20 %. The results illustrated that the accuracy of the ANN method is higher than applied CFD method considering relative error and consumed time and price.

The ANN model was created in research to forecast how well solar collectors with TES materials would function. The built ANN model has two parameters in the output layer (heating capacity and thermal efficiency) and five parameters in the input layer (ambient temperature, solar radiation intensity, incidence angle, slope angle, and azimuth angle). The ANN structure was modeled using the logistic sigmoid transfer function and the BP learning method. Ten neurons with two layers for the hidden layer was the ideal number for precise data prediction, based on the tests that were conducted. The RMSE, COV, and R^2 values for the anticipated heating capacity data were 7840.56, 0.6204, and 0.9995, respectively. Additionally, the efficiency values were 0.0075, 0.3384, and 0.9999, respectively, indicating that the ANN model was performing as intended [197]. Other studies, however, have only focused on forecasting the storage fluid's behavior in thermal systems. The ANN approach was applied in a study to forecast an energy storage tank's performance. Various networks were assessed in order to determine the system's performance and outlet temperature. The optimal structure was chosen based on repeatability and lower error compared to laboratory data after each network was analyzed independently using 32 distinct structures for two energy storage and release modes [198]. The ANN approach with various architectures was utilized in a study to forecast the behavior of the PCM and working fluid in a photovoltaic-thermal system. The findings demonstrated that the SOFM 4 system's electrical and thermal efficiency was increased by the introduction of PCM. The system can be assessed using the chosen structure (self-organizing maps) with the proper accuracy ($R^2 < 0.94$), according to Al-Waeli et al. [199].

Research by Barghi Jahromi et al. [9] conducted an experimental thermo-economic analysis of an indirect cabinet solar drier employing evacuated tube collectors and PCM for drying Jerusalem artichoke slices, yielding numerous notable findings. The researchers noted that PCM significantly influenced the removal of water content from 5 mm thick slices of Jerusalem artichoke ($P < 0.05$), but elevated airflow speeds decreased drying time, albeit not considerably ($P > 0.05$). The

incorporation of PCM increased the activation energy to 33.4 kJ/mol and enhanced the total drying efficiency by 1.51–7.81 %, they also demonstrated. According to their analysis, the installation of PCM reduced specific energy consumption from 14.51 to 13.38 MJ/kg, and during the drying process, exergy efficiency varied between 35.3 % and 59.7 %. The research identified enhanced system parameters that decreased the payback period to 22 months, indicating a 15 % enhancement compared to baseline settings. They utilized artificial neural networks, CFD, and EPR to forecast the system's thermal behavior, with the neural network method demonstrating reduced RMSE and COV values and enhanced R^2 in comparison to CFD. The EPR method accurately predicted outlet temperature and collector thermal efficiency, achieving a high R^2 value exceeding 0.98. They established that the ideal working condition, including comprehensive exergetic, economic, and quality factors, involved an airflow rate of 0.9 kg/s with PCM assistance (Fig. 32).

They used nano-enhanced phase transition material (AlO₃-Paraffin wax) in their experimental study of a concentrated photovoltaic-thermal solar dryer (CPV/TSD), Karaağaç et al. [44] carried out a thorough thermodynamic analysis based on the first and second laws of thermodynamics. While demonstrating that the thermal energy transferred to the nano-enhanced PCM successfully maintained greenhouse temperatures for the first 100 min, despite significant decreases in ambient solar radiation, the researchers discovered that the CPV/TSD system had an exergy efficiency of 8 % and an overall thermal energy efficiency of 20 %. Their method successfully decreased the mushrooms' MC from 17.45 g water/g dry matter to 0.0516 g water/g dry matter, giving the system a drying rate of 0.437 g matter/g dry matter min. In order to predict crucial drying parameters like MC and moisture ratio, the study used two machine learning algorithms: ANN and SVM. The predictive performance of these algorithms was evaluated using four distinct evaluation metrics (R^2 , RMSE, MBE, and MAE). Karaağaç et al. [44] found that both ANN and SVM algorithms exhibited commendable predictive performance; however, they determined that the ANN approach outperformed SVM across all evaluation metrics, rendering it the preferred method for predicting drying parameters in this specific solar drying application employing nano-enhanced PCM (Fig. 33).

Arunkumar et al. [10] conducted a study examining the use of five different machine learning algorithms to predict performance parameters in red chilli drying operations within an indirect solar dryer system. The findings underscore the potential of machine learning models as a viable alternative to conventional modeling methods for improving drying efficiency through precise system predictions. The experimental design included three distinct Energy Storage Element (ESE) configurations: pure paraffin wax (sample 1), glass pieces combined with paraffin wax (sample 2), and crushed glass pieces mixed with paraffin wax (sample 3). Additionally, three specific airflow rates (0.3 ms^{-1} , 0.4 ms^{-1} , and 0.5 ms^{-1}) were examined to assess performance variations comprehensively. Arunkumar et al. [10] examined various operational parameters such as solar intensity, airflow rate, ESE configuration type, Ambient Temperature (AT), Collector Inlet Temperature (CIT), Collector Outlet Temperature (COT), Drying Chamber Inlet Temperature (DCIT), and Drying Chamber Outlet Temperature (DCOT) to assess their effects on key performance indicators, namely MC and Collector Efficiency (CE). The experimental results indicated that the solar dryer employing sample 3 ESE (crushed glass pieces with paraffin wax) at the maximum airflow rate of 0.5 ms^{-1} attained optimal performance, achieving the target final MC in only 25 h. Arunkumar et al. [10] evaluated various ML techniques and determined that the Gradient Boosting Regression Tree (GBRT) algorithm demonstrated exceptional predictive performance for MC, achieving an R^2 value of 0.999964 and minimal error metrics (MAE of 0.001303, RMSE of 0.001882, MSE of 0.000004). Additionally, it showed strong predictive capabilities for collector efficiency, with an R^2 value of 0.994064 and error metrics (MSE of 0.000109, MAE of 0.007413, RMSE of 0.010442). The analysis indicated that the solar collector outlet temperature is the most significant factor influencing

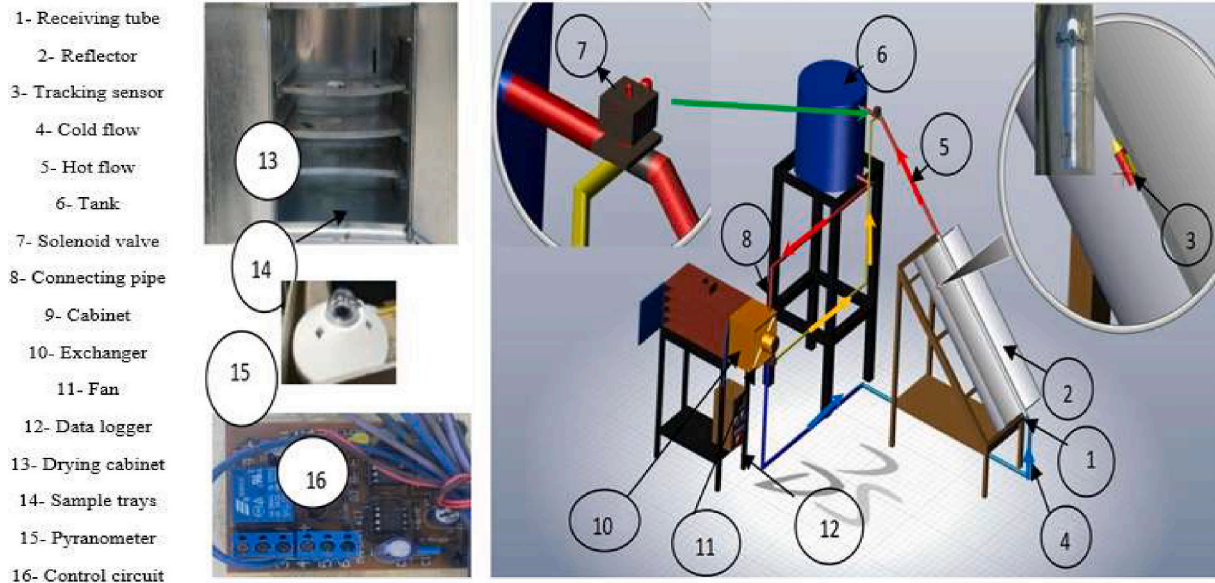


Fig. 31. Schematic of a solar dryer equipped with a PTSC and heat storage in a spiral copper tube in the tank [196].

both the MC of red chilli and the overall efficiency of the drying system. The researchers concluded that both the ESE configuration and airflow rate have a substantial impact on the agricultural drying process, while also highlighting the effectiveness of machine learning methods in predicting performance in solar drying applications (Fig. 34).

Sehrawat et al. [200] investigated significant performance limitations of greenhouse dryers, specifically heat losses from the north wall and limited operating hours. They developed and assessed a modified hybrid greenhouse drying system that integrates TES and bifacial photovoltaic thermal (BIFPVT) technology for concurrent electrical and thermal energy generation. The researchers wrapped thermocol in aluminum foil to create a reflective surface on the north wall, thereby minimizing heat losses from this traditionally problematic area. Sehrawat et al. [200] improved the system's sustainability by integrating PCM that stored thermal energy during daylight, allowing for ongoing drying operations during non-sunshine periods and thus extending the system's functional capacity. Experimental measurements indicated that the BIFPVT component produced electrical power between 2.0 and 85.5 W from the front side and 0.6 to 81 W from the rear side, corresponding to

solar radiation intensities of 34–950 W/m² and 13–350 W/m² on the front (IF) and rear (IR) surfaces, respectively. The thermal performance analysis by Sehrawat et al. [200] indicated that the drying chamber maintained temperatures 16–44 % above ambient conditions on both test days, confirming the efficacy of their design modifications in improving thermal retention and energy utilization. Sehrawat et al. [200] utilized artificial neural network modeling alongside experimental data to enhance their understanding of system behavior and optimize performance parameters. They subsequently applied multi-objective genetic algorithm tools to determine optimal operating configurations, thereby creating a thorough methodology for analyzing and improving the performance of their energy-efficient hybrid greenhouse drying system (Fig. 35).

A comprehensive study of the effects of PCM on cabinet dryer performance was carried out by Mirzaee et al. [16], with an emphasis on thermal and drying efficiency across three distinct PCM placement configurations: bottom, middle, and top trays. According to the experimental findings, drying tomato slices required a specific energy consumption ranging from 11.32 to 14.87 MJ/kg. Compared to other

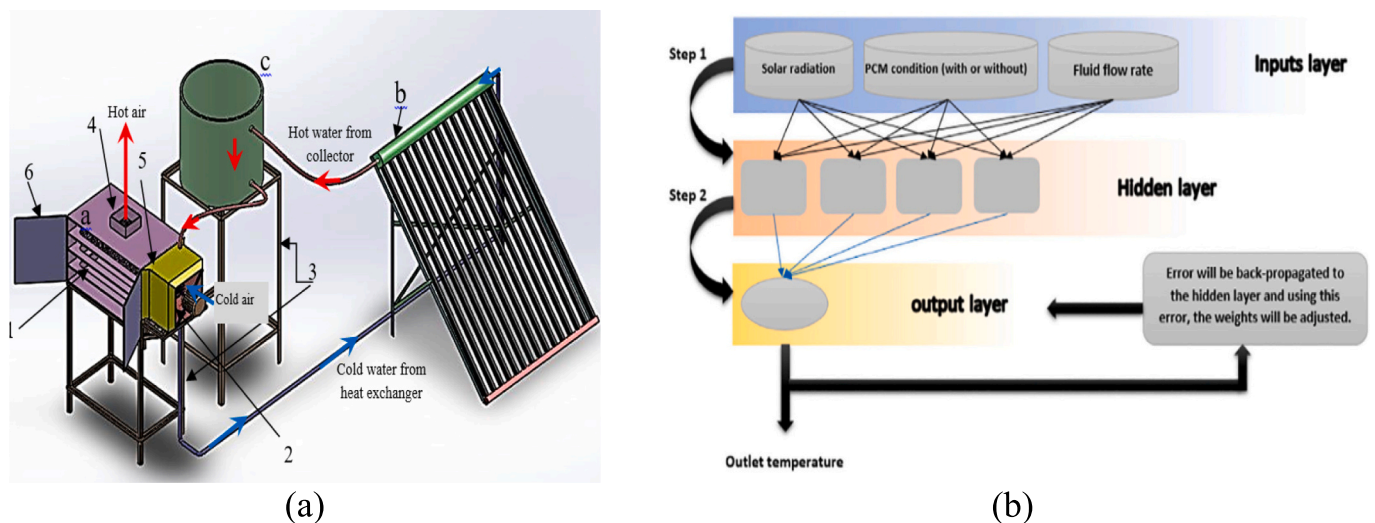


Fig. 32. (a) Cabinet-based indirect solar dryer system connected to ETSC and using PCM melted inside a U-shaped copper tube located in the water tank, (b) artificial neural network structure used in solar dryer system with thermal storage [9].

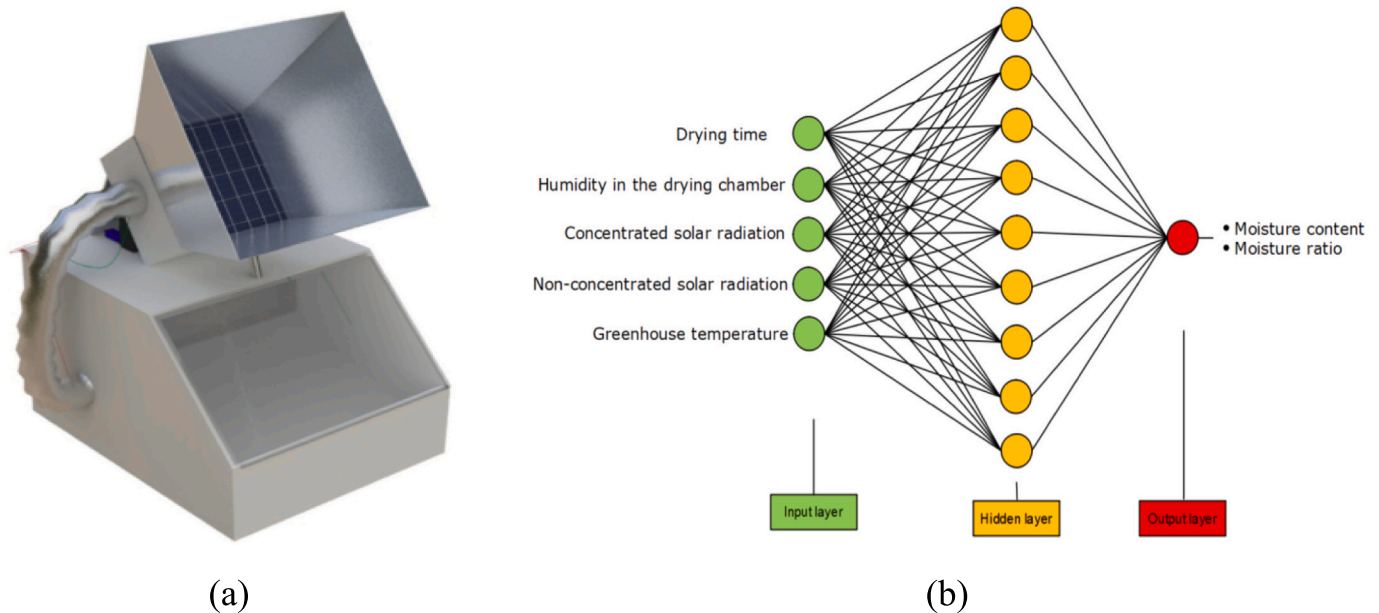


Fig. 33. (a) Schematic of a concentrated photovoltaic-thermal solar dryer equipped with PCM on the iron plate under the photovoltaic panel, (b) ANN structure for the concentrated photovoltaic-thermal solar dryer system [44].

setups, drying durations were noticeably reduced when PCM was placed on the lowest tray. Additionally, the overall thermal efficiency of the cabinet drying system varied between 35.23 % and 38.92 %. Their experimental work was enhanced by CFD simulations, which accurately predicted the thermal performance of the cabinet dryer ($R^2 > 0.95$). Additionally, they utilized EPR methodology to forecast the output temperature and efficiency parameters of the cabinet drying system. The researchers performed an environmental impact assessment utilizing Life Cycle Assessment (LCA) methodology to compare their solar drying system with conventional thermal electric dryers. Their analysis indicated significant differences in environmental impacts between the two technologies across all assessed categories. This multifaceted approach, which integrates experimental testing, computational modeling, and environmental assessment, yields valuable insights into optimizing PCM implementation in cabinet dryers. It highlights the performance advantages and environmental benefits of solar drying technologies enhanced with strategically positioned PCM, in contrast to conventional

electric drying systems. Additionally, it establishes predictive models that can aid in the further optimization and development of sustainable drying technologies (Fig. 36).

Ebrahimi et al. [48] examined the influence of PCM on the thermal performance and drying efficiency of FPC, focusing on the ongoing issue of low thermal efficiency in solar collector applications across multiple domains. The study analyzed four different PCM positions within the collector structure, evaluating performance metrics in comparison to a baseline collector lacking PCM integration. The experimental results indicated that the specific energy consumption for drying tomato slices varied from 9.01 to 11.12 MJ/kg, and the placement of PCM at the collector's end section notably decreased drying time by about 21.87 %. The study revealed that enhancements in thermal efficiency ranged from 5.02 % to 10.13 % based on the positioning of PCM, whereas the overall system efficiency varied from 21.92 % to 25.72 % across the tested PCM configurations. The researchers performed CFD simulations utilizing ANSYS software, attaining high accuracy in forecasting the collector's

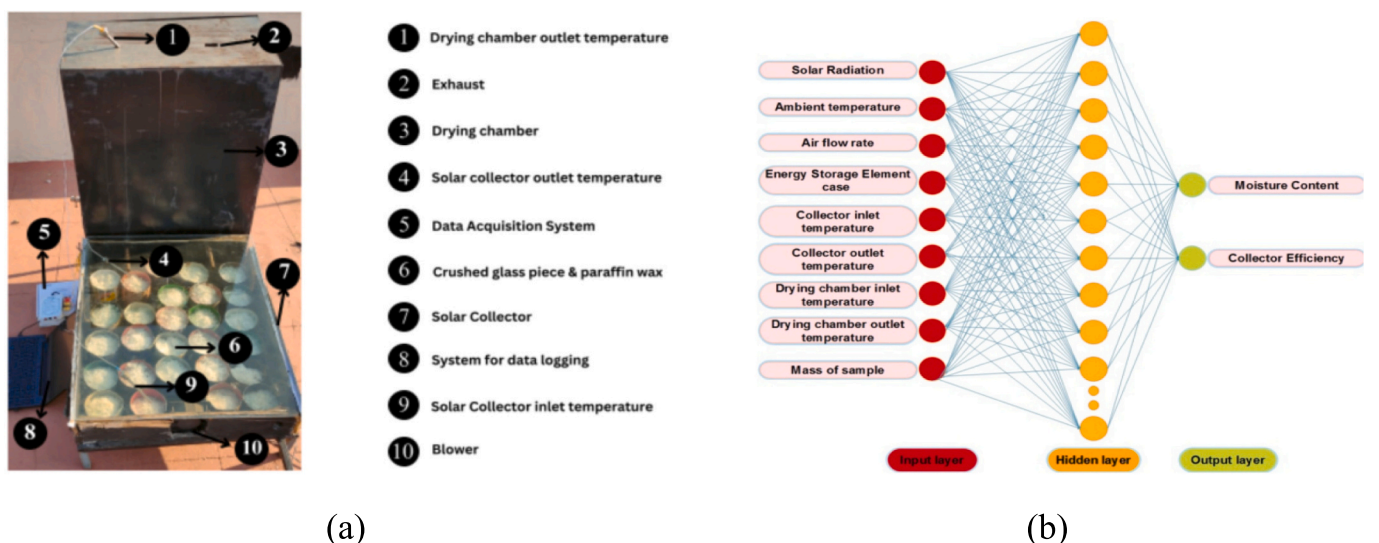


Fig. 34. (a) Indirect solar dryer with PCM in solar air heater, (b) ANN structure for solar dryer system with PCM [10].

thermal performance ($R^2 > 0.9432$). They also applied evolutionary polynomial regression modeling for each PCM position, which effectively predicted outlet temperature and thermal efficiency with remarkable precision ($R^2 > 0.98$). The study concluded that placing PCM at the terminal section of the collector resulted in optimal performance characteristics, such as more uniform drying, improved temperature regulation, reduced drying times, and superior quality of dried tomato slices compared to other PCM placements. This finding establishes important design parameters for enhancing solar collector efficiency through strategic PCM integration (Fig. 37).

The application of thin-layer drying models for agricultural goods in solar drying systems was thoroughly reviewed Kidane et al. [201]. This study specifically analyzes the application of thin-layer models in solar dryers, contrasting with existing reviews that broadly address various drying methods. This review offers a bibliometric analysis indicating that research in this field commenced in 1976 and has experienced substantial growth, peaking in 2021, with India and China as the primary contributors. The findings underscore the significance of variables like inlet air temperature and pretreatments in enhancing the drying process. The choice of a suitable drying model is contingent upon the distinct characteristics of the agricultural product, the particular drying conditions, and the model's predictive precision across different environmental variables. The Midilli et al. model is notable for its efficacy across a range of products. While suggesting future research possibilities for the application of ANNs in this domain, the review emphasizes the growing importance of ANNs in improving the prediction of drying behavior for agricultural products utilizing solar drying technologies [201].

Gupta et al. [43] introduced a mixed-mode solar dryer that incorporates a semi-transparent photovoltaic thermal (PVT) collector to evaluate drying and thermal performance using both computational and experimental methods. The research optimized airflow by evaluating three inlet velocities (0.048, 0.096, and 0.144 m/s). The results indicated that Case 2 (0.096 m/s) achieved the most uniform temperature distribution, whereas Case 3 (0.144 m/s) exhibited the highest thermal efficiency and enhanced fluid flow characteristics. Thermal efficiencies of 33.20 % and 31.14 % were achieved for Case 3, both numerically and experimentally, in contrast to the lower values observed in Cases 1 and 2. Case 3 demonstrated notable enhancements in Reynolds and Nusselt numbers compared to Case 1. The experimental results aligned closely with CFD simulations, exhibiting deviations between 2.24 % and 4.90 %. The study employed ANN models with multiple hidden layers to predict drying performance parameters and determine optimal drying conditions.

7. Challenges

The development and optimization of solar dryers utilizing PCM and advanced neural algorithms encounter several significant challenges. The accurate acquisition and monitoring of environmental and operational parameters, including ambient temperature, solar radiation, airflow, and humidity, constitutes a primary obstacle. Experimental setups frequently encounter measurement errors and systematic biases resulting from sensor limitations and variable environmental conditions. Inaccuracies can undermine data quality, hindering the acquisition of reliable datasets for empirical analysis and model training.

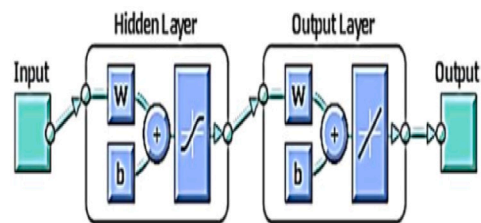
A notable challenge stems from the inherent complexity of solar drying systems, particularly when integrating PCM and advanced control techniques. The placement, type, and quantity of PCM should be customized for each application, considering the properties of the dried product, the design of the drying chamber, and the characteristics of the solar collector. Attaining uniform temperature distribution and effective



Fig. 36. Solar dryer system with PCM located inside the cabinet in an encapsulated form [16].



(a)



(b)

Fig. 35. (a) Eco-friendly greenhouse drying system for drying various agricultural products, (b) ANN algorithm structure for greenhouse dryer [200].

thermal storage or release presents significant technical challenges, especially as systems transition from laboratory prototypes to commercial applications.

The incorporation of neural algorithms with evolutionary polynomial regression adds further complexities. These models demonstrate enhanced predictive accuracy relative to traditional methods; however, their effectiveness is significantly influenced by the quality and diversity of the training data. Over-fitting, restricted generalizability to novel operating conditions, and the computational resources necessary for real-time prediction and control may impede practical implementation. The effective implementation of these models in practical environments necessitates thorough validation, consistent updates, and smooth integration with sensor networks and control systems. Coordinated efforts are essential to fully realize the potential of solar dryers utilizing PCM and neural algorithms, thereby advancing sustainable, efficient, and resilient food preservation technologies.

Beyond technical and modeling aspects, several practical implementation challenges remain critical for real-world deployment. First, the relatively high cost and limited local availability of PCM, sensors, and advanced control units can hinder large-scale adoption, particularly in resource-constrained agricultural settings. Second, scaling up solar dryers from laboratory prototypes to commercial or industrial levels introduces additional complexity, such as ensuring structural integrity, maintaining uniform thermal distribution, preventing material degradation, and minimizing long-term maintenance costs. Third, comprehensive field validation is essential: while most systems perform well under controlled laboratory conditions, their effectiveness must be demonstrated across diverse climates, products, and operational environments to confirm durability, cost-effectiveness, and robustness. Moreover, user acceptance and infrastructure requirements present further obstacles, farmers and small-scale producers may face difficulties in adopting systems that rely on advanced sensor networks, stable power supply, or internet connectivity. Finally, sustainability concerns, including the lifecycle performance, recyclability, and replacement frequency of PCM, must be addressed to ensure that such systems remain environmentally and economically viable. Addressing these multifaceted challenges will be decisive in translating PCM-based solar drying technologies integrated with intelligent optimization models into scalable, sustainable, and widely accepted food preservation solutions.

8. Conclusion

This review thoroughly analyzes the integration of neural algorithms and EPR in optimizing and assessing solar drying systems that utilize PCM. The rising demand for sustainable and energy-efficient food preservation methods underscores the significance of solar dryers, which provide notable benefits compared to traditional drying techniques, such as decreased energy usage, lower operational expenses, and reduced environmental impact. The integration of PCM in solar dryers enhances system performance through improved temperature uniformity, prolonged drying durations beyond daylight hours, and better quality and shelf life of dried products. Recent studies on solar dryers with PCM have reported significant gains in energy storage and utilization efficiency. Such enhancements directly contribute to improved product quality, reduced processing time, and increased adaptability under variable weather conditions.

Neural algorithms, including ANNs, support vector machines, decision trees, and advanced ensemble methods, exhibit enhanced predictive accuracy and modeling capabilities relative to conventional mathematical models. These advanced methods facilitate the precise estimation and optimization of essential parameters, including MC, drying efficiency, and energy consumption, even in complex and variable operating conditions. EPR has demonstrated effectiveness in predicting critical performance metrics such as collector outlet temperature and system thermal efficiency, frequently surpassing traditional CFD models in accuracy and computational efficiency.

Notwithstanding these advancements, numerous challenges persist. Data acquisition that is both accurate and reliable is crucial for experimental validation and model training; however, it is frequently impeded by sensor limitations and environmental variability. The integration of PCM and advanced AI models into various dryer designs poses significant technical and practical challenges, particularly in the context of scaling for commercial applications. Additionally, thorough environmental and economic evaluations are essential to grasp the long-term sustainability and cost-effectiveness of these systems.

The integration of PCM and neural algorithm-based optimization signifies a significant advancement in solar drying technologies. Ongoing research and development in this domain should prioritize enhancements in data quality, investigation of innovative AI architectures, optimization of PCM materials and configurations, and execution of rigorous field demonstrations. Addressing these areas allows solar dryers to significantly reduce post-harvest losses, enhance food security, and facilitate the transition to sustainable agricultural practices.



Fig. 37. Solar dryer system with PCM placed inside a spiral copper tube inside the absorber plate [48].

9. Future work

Future research in solar drying systems, especially those utilizing PCM and advanced Machine learning optimization, presents promising avenues to overcome current limitations and enhance the functionalities of these technologies. The development of more accurate and robust data acquisition systems is a critical area for future research. Enhancing the precision and reliability of environmental and operational data, including temperature, humidity, solar radiation, and airflow, will markedly improve the quality of experimental results and the efficacy of AI-based modeling and optimization. Advancements in sensor technology, data fusion, and real-time monitoring are critical for minimizing systematic errors and maintaining the integrity of datasets utilized in simulation and control.

A significant direction for future research involves the integration and comparative analysis of a wider array of neural algorithms and machine learning techniques. Artificial neural networks, decision trees, and evolutionary polynomial regression have demonstrated significant efficacy; however, there exists substantial potential in investigating advanced architectures, including deep learning models, RNN, LSTM networks, and ensemble methods. These methods may enhance the accuracy, generalizability, and adaptability of predictive models for solar drying systems across various operating conditions and agricultural products.

The optimization of PCM selection, placement, and system integration is a critical area of research. Future research should concentrate on discovering new PCM that exhibit superior thermal properties, increased durability, and enhanced compatibility with diverse solar dryer configurations. Examining novel approaches for the encapsulation, storage, and distribution of PCM within the dryer chamber may result in enhanced uniformity of temperature profiles and improved energy storage efficiency. The development of hybrid systems that integrate sensible and latent heat storage, or combine various types of solar collectors, presents opportunities to enhance drying performance and operational flexibility.

The transition from laboratory-scale research to real-world applications necessitates pilot projects and field demonstrations that focus on scalability, system robustness, and user adoption. Collaboration among researchers, industry stakeholders, and policymakers is crucial for enabling technology transfer, establishing best practice guidelines, and encouraging the broad adoption of intelligent, energy-efficient solar drying solutions. Finally, a research roadmap for the future integration of PCM and AI in solar drying systems is proposed, highlighting the following directions:

- **Advanced PCM Materials and Configurations:** Development of novel PCMs and optimized placement strategies to improve thermal management and energy efficiency.
- **Real-Time AI-Based Control:** Deployment of neural networks, ensemble models, and reinforcement learning for dynamic control of drying parameters.
- **Hybrid Optimization Methods:** Combining AI with conventional simulation techniques, such as CFD and energy-exergy modeling, for predictive and adaptive operation.
- **Scalable Industrial Applications:** Design and demonstration of pilot- and industrial-scale PCM-AI integrated dryers, with attention to economic feasibility and environmental impact.
- **Data-Driven Model Enhancement:** Establishing large experimental datasets to train AI models for improved accuracy, robustness, and generalization.

Furthermore, specific recommendations for real-world integration have been added to guide practical implementation:

- **Implement pilot-scale PCM-integrated solar dryers** with AI-based control in actual agricultural settings to assess performance, reliability, and user acceptance.
- **System design and PCM selection:** These PCMs should be designed to directly match the solar radiation levels and product spectrum of each region. For example, hydrated salts can be used as thermal storage for humid regions, while paraffin-based materials are more suitable for regions where the product is processed at high temperatures. Standard and scalable encapsulated units (for tubular heat exchangers or plate panels) should be designed and manufactured. These modules should be capable of rapid installation in conventional dryers and solar heating systems, making them more cost-effective by reducing initial and maintenance costs.
- **Develop predictive and adaptive AI algorithms** that dynamically adjust dryer operation based on real-time environmental conditions and PCM thermal behavior.
- **Conduct economic and environmental assessments** to evaluate cost-effectiveness, energy savings, and sustainability, guiding scalable adoption.
- **Establish robust sensor networks and standardized datasets** to train AI models for diverse crops, dryer types, and climatic conditions.
- **Collaborate with farmers, manufacturers, and policymakers** to design user-friendly systems and establish best practice guidelines.

Addressing these future research directions will enable the field to better realize the potential of solar dryers in promoting sustainable agriculture, minimizing post-harvest losses, and enhancing global food security.

CRediT authorship contribution statement

Mohammad Saleh Barghi Jahromi: Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ayla Sayedolasgari:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **S. Madhankumar:** Writing – review & editing, Visualization, Conceptualization. **Hadi Samimi Akhijahani:** Writing – review & editing, Visualization, Conceptualization. **Payman Salami:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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